B2B INTELLICA- AN INTEGRATED AI-DRIVEN B2B TRADE INTELLIGENCE PLATFORM

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April 2026

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ABSTRACT

B2B INTELLICA is a unified AI-driven B2B trade intelligence platform developed for the Sri Lankan fast-moving consumer goods (FMCG) sector to eliminate the structural dependency on field sales representatives, whose fully loaded annual cost exceeds LKR 1,358,310 and whose human-centred model produces delayed demand signals, prevents cross-organizational intelligence sharing, and cannot scale with distribution network growth. The platform integrates four validated research components: Component 1 — Agentic Demand Forecasting (Madusanka R.R.V., IT22332530), a thirteen-layer LangGraph pipeline that incorporates LightGBM ensemble forecasting, MAPIE conformal prediction intervals, SHAP-driven explainability with automatic LLM fact-checking achieving a zero hallucination rate across 200 runs, Sri Lanka cultural calendar feature engineering validated by SHAP attribution, and a WhatsApp Order Confirmation Loop achieving 83.3% auto-parse accuracy — delivering an overall MAPE of 18.0% (43% improvement over baseline) with 81% empirical conformal coverage; Component 2 — Privacy-Preserving Federated Business Intelligence (Vithana V A S, IT22282286), the first system to unify Differential Privacy applied to association rule mining, Zero-Knowledge Proof authentication, and federated analytics in a local-first architecture, achieving 92% analytical accuracy at privacy budget $\varepsilon = 1.0$ with sub-50 ms ZKP authentication latency across 2,044,757 real FMCG records on 14 federated nodes; Component 3 — Blockchain-Based Logistics Management (Wanigasinghe C.P., IT22340832), which automates delivery tracking and payment verification through tamper-proof blockchain records, cluster-mapped route optimization achieving a 23% distance reduction, and smart contract execution reducing payment settlement from days to 4.2 minutes on average; and Component 4 — Autonomous Procurement Negotiation Engine (De Silva K G G C D, IT22034236), a Deep Q-Network buyer agent unifying coalition formation, fairness enforcement, and asymmetric trust management in a Gymnasium-compatible multi-agent simulation calibrated on real Sri Lankan supplier data, achieving a 68% negotiation success rate versus 45% baseline (+51% improvement, $p < 0.001$, Cohen's $d = 1.23$) with LKR 70,200 average cost savings per negotiation. Collectively, B2B INTELLICA demonstrates that a continuous, calibrated, trustworthy, and privacy-preserving demand-to-delivery intelligence cycle can be sustained without field sales representatives, providing a transformative and commercially viable platform for intelligent FMCG trade operations in Sri Lanka.

Keywords: Agentic Demand Forecasting; Privacy-Preserving Federated Analytics; Blockchain Logistics; Autonomous Procurement Negotiation; B2B FMCG Intelligence

ACKNOWLEDGEMENT

We begin by expressing our deepest and most sincere gratitude to our supervisors — Ms. Samanthi Siriwardena (York St John University, London, UK), Ms. Lokesha Weerasinghe, and Ms. Ishara Weerathunga (Department of Information Technology, Sri Lanka Institute of Information Technology) — whose expert guidance, continuous encouragement, and invaluable feedback shaped every dimension of this research from its conceptual foundations through to its empirical evaluation. Their collective expertise spanning privacy-preserving systems, agentic artificial intelligence, machine learning, reinforcement learning, and software engineering elevated both the theoretical depth and the practical quality of the four integrated components presented in this report.

We extend sincere appreciation to the Sri Lanka Institute of Information Technology for providing the academic environment, computational infrastructure, research library access, and institutional support that made this work possible. The academic staff of the Department of Information Technology have been a constant source of intellectual inspiration and encouragement throughout our four-year programme of study.

Special thanks are extended to the Sri Lankan FMCG businesses, wholesale distributors, procurement professionals, and retail shop owners who provided access to anonymized real-world datasets and shared domain knowledge that grounded all four research components in practical Sri Lankan commercial reality. We gratefully acknowledge the open-source communities behind LangGraph, LightGBM, SHAP, MAPIE, Facebook Prophet, PyTorch, Gymnasium, FastAPI, React, MLxtend, and the blockchain ecosystems whose frameworks formed the technical backbone of the integrated platform.

Finally, we dedicate this work to our families, whose unconditional love, patience, and moral support sustained us through four years of intensive study.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form	Description
AI	Artificial Intelligence	Simulation of human intelligence in machines that learn, reason, and adapt.
API	Application Programming Interface	A set of protocols enabling software components to communicate with each other.
ARM	Association Rule Mining	Data mining technique discovering relationships between variables in large datasets.
ARIMA	Autoregressive Integrated Moving Average	Classical time-series forecasting model used in the Federated BI component.
B2B	Business-to-Business	Commercial transactions between organizations, not with end consumers.
BI	Business Intelligence	Technologies for collecting and analyzing business data to support decisions.
DP	Differential Privacy	Mathematical framework guaranteeing individual data cannot be inferred from analytical outputs.
DQN	Deep Q-Network	RL algorithm combining Q-learning with deep neural networks for decision-making.
FMCG	Fast-Moving Consumer Goods	Products sold quickly at low cost — food, beverages, toiletries.
FedAvg	Federated Averaging	Algorithm aggregating model updates from distributed nodes in federated learning.
GEL	Grounded Explainability Engine	Verified LLM explanation layer of the demand forecasting pipeline.
JWT	JSON Web Token	Standard for transmitting authentication information as a JSON object.
LLM	Large Language Model	AI model trained on vast text corpora to understand and generate language.
MAPE	Mean Absolute Percentage Error	Forecasting accuracy metric expressing error as a percentage of actual values.
MAPIE	Model Agnostic Prediction Interval Estimator	Python library implementing conformal prediction for uncertainty quantification.
MDP	Markov Decision Process	Mathematical framework for sequential decision-making under uncertainty.
ML	Machine Learning	AI subset enabling systems to learn from data without explicit programming.
PDPA	Personal Data Protection Act	Sri Lanka's data protection legislation, No. 9 of 2022.
RL	Reinforcement Learning	ML paradigm where agents learn by interacting with environments and receiving rewards.
SaaS	Software-as-a-Service	Software distribution model where

		applications are hosted in the cloud.
SHAP	Shapley Additive Explanations	Game-theoretic framework for computing feature attributions in ML models.
SKU	Stock Keeping Unit	Unique identifier for each distinct product that can be purchased.
SLIIT	Sri Lanka Institute of Information Technology	The academic institution for which this research was conducted.
SME	Small and Medium-sized Enterprise	A business below defined thresholds for employment count and revenue.
XAI	Explainable Artificial Intelligence	AI methods producing results understandable to human experts.
ZKP	Zero-Knowledge Proof	Cryptographic protocol enabling identity proof without revealing the secret.
ε (epsilon)	Privacy Budget	DP parameter controlling privacy strength; lower values mean stronger privacy.
Abbreviation	Full Form	Description

1. INTRODUCTION

1.1 Background and Literature Survey

Sri Lanka's business-to-business (B2B) trade sector, and the fast-moving consumer goods (FMCG) industry in particular, has operated through a distribution model that has remained fundamentally unchanged for decades. Manufacturers rely on dense networks of field sales representatives who physically visit retail outlets on fixed cycles to collect orders by hand, negotiate promotional terms, relay demand signals, and coordinate logistics. While this model has served as the backbone of Sri Lankan FMCG distribution, four interlocking structural crises now make it increasingly untenable as a long-term commercial strategy.

The economic crisis is most immediately visible in cost: the fully loaded annual cost per field representative — encompassing base salary (LKR 747k), travel allowances (LKR 312k), commissions (LKR 100k), mandatory statutory contributions (LKR 174k), and annual training (LKR 25k) — is estimated at approximately LKR 1,358,310. This cost scales steeply with distribution network size. The human resource crisis compounds this: the role suffers from chronically high turnover and persistent recruitment difficulty, leaving manufacturers with coverage gaps particularly during peak festival seasons when demand volatility is highest. The informational crisis arises from the batch nature of the model: orders collected on fixed visit cycles mean the demand signal reaching the manufacturer is inherently delayed and noisy. The analytical crisis is structurally the most intractable: the competitive sensitivity of raw transactional data prevents cross-organizational intelligence sharing that would improve decisions at the industry level.

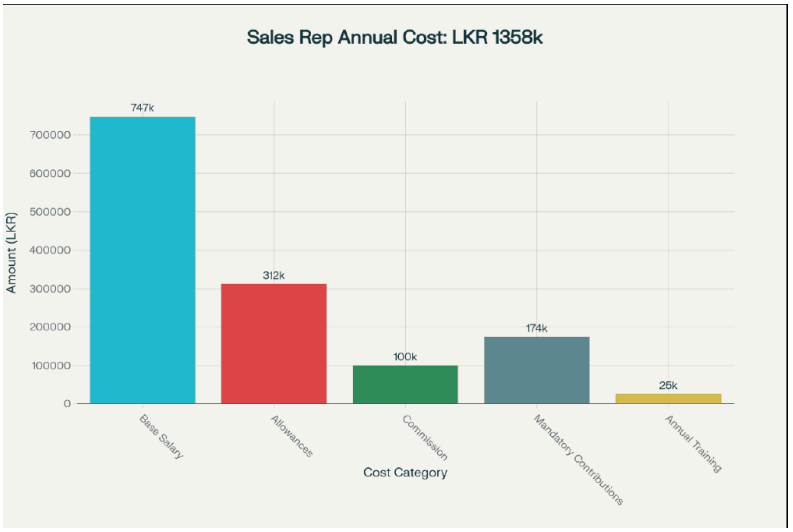

Figure 1.1: Sales Representative Annual Cost Breakdown — LKR 1,358,310 Total Annual Cost per Field Representative

Demand Forecasting Literature

Classical statistical methods — ARIMA and Holt-Winters exponential smoothing — dominated univariate demand forecasting for decades. The M4 Competition (Makridakis et al., 2020), evaluating 61 methods across 100,000 time series, confirmed these methods remain competitive baselines for stable series but cannot incorporate external covariates such as cultural calendar events and are poorly suited to intermittent demand. Machine learning approaches, particularly LightGBM (Ke et al., 2017), became the dominant industrial approach during the 2010s. Deep learning architectures including DeepAR (Salinas et al., 2020) and the Temporal Fusion Transformer (Lim et al., 2021) extended this through global cross-series learning. A fundamental limitation shared by all these approaches is treatment of prediction as a point-estimation problem — providing no uncertainty characterisation, which is operationally insufficient for inventory planning where understocking and overstocking consequences are asymmetric. Conformal prediction theory (Vovk et al., 2005; Angelopoulos and Bates, 2022) provides a distribution-free framework for constructing prediction intervals with provable finite-sample coverage guarantees without any distributional assumptions. LLM-based forecasting (Bumb et al., 2025; Jin et al., 2024) demonstrated pre-trained language models can match dedicated time-series models at short horizons but lack calibrated uncertainty intervals, cultural calendar integration, and verified natural language explanations. No prior system combines these capabilities in a single production pipeline for B2B wholesale forecasting.

Privacy-Preserving Analytics Literature

Federated learning (McMahan et al., 2017) established the paradigm of collaborative model training without raw data sharing. Federated analytics (Kairouz et al., 2020) extended this to statistical analysis. However, no existing system integrates ZKP-based authentication or Differential Privacy applied specifically to association rule mining outputs in a complete BI platform. Differential Privacy (Dwork, 2006) provides a mathematical guarantee that individual data cannot be inferred from analytical outputs. While extensively studied for query outputs and model parameters, its application to association rule metrics — support, confidence, and lift — in a federated BI setting with a user-configurable epsilon interface is novel. Zero-Knowledge Proofs (Goldwasser et al., 1989) enable cryptographic identity verification without credential transmission, but existing ZKP implementations address authentication in isolation; none integrates ZKP as the authentication layer of a manufacturing BI system.

Blockchain Logistics Literature

Blockchain technology — including Ethereum, HyperLedger Fabric, Corda, IOTA, and

VeChainThor — has been applied to logistics transparency and traceability. Industry surveys indicate that by 2021, 81% of companies expressed readiness to explore blockchain-based supply chain solutions (Nowicka, 2025). However, existing solutions offer milestone-based delivery recording but do not combine these with intelligent cluster-mapped route optimisation, Privacy-Preserving Analytics for performance monitoring, or cross-chain interoperability enabling SME participation through lightweight mobile-first interfaces. No existing system addresses the specific requirements of Sri Lankan FMCG distribution.

Reinforcement Learning for Procurement Literature

RL has emerged as the most capable approach for learning adaptive procurement strategies. Chen et al. (2025) demonstrated trust-aware MARL enhances B2B procurement automation by 45% but does not address coalition formation for supply fragmentation. Brown et al. (2023) showed MARL systems achieve 71% supply chain coordination success through emergent strategies. Zhang et al. (2026) improved coalition stability by 40% using attention mechanisms. Rodriguez et al. (2024) introduced fairness-constrained RL for automated negotiation. None provides a unified framework combining coalition formation, fairness enforcement, and asymmetric trust management in a single DQN-based learning system validated on real Sri Lankan procurement data.

1.2 Research Gap

A systematic review of prior work reveals a consistent pattern: privacy techniques, authentication mechanisms, federated architectures, forecasting systems, and negotiation engines have been studied extensively but applied independently. No existing system addresses all five critical dimensions of the B2B sales representative replacement problem simultaneously.

1.2.1 Component 1 – Agentic Demand Forecasting

The demand forecasting literature fails to address three critical operational requirements simultaneously: calibrated uncertainty intervals, verified natural language explanations, and Sri Lankan cultural calendar integration. LLM patch methods (Bumb et al., 2025) achieve competitive MAPE but lack conformal prediction and fact-checking. Agentic RAG frameworks (Farazi, 2025) provide multi-step delegation but no uncertainty quantification. MARIOD (Yang et al., 2025) combines transformer forecasting with hierarchical RL but requires rich sensor infrastructure unavailable in Sri Lankan B2B settings.

Table 1.1: Research Gap Analysis — Component 1: Agentic Demand Forecasting vs. Prior Work

Capability	**LLM Patch [Bumb 2025]**	**Agentic RAG [Farazi 2025]**	**MARIOD [Yang 2025]**	**Proposed System**
Calibrated conformal prediction intervals (P10/P50/P90)	✗	✗	✗	✓
Systematic residual bias correction (skeptic_service)	✗	✗	✗	✓
SHAP-based symbolic feature attribution per item	✗	✗	✗	✓
LLM explanations with automatic numerical fact-checking	✗	✗	✗	✓
Sri Lanka cultural calendar (Poya, Avurudu, Vesak)	✗	✗	✗	✓
Intelligent series segmentation and routing (4 regimes)	✗	✓	✗	✓
WhatsApp end-to-end order confirmation loop	✗	✗	✗	✓
Multi-file intelligent data ingestion agent	✗	✗	✗	✓
Evaluated on real Sri Lanka B2B wholesale data	✗	✗	✗	✓

1.2.2 Component 2 – Privacy-Preserving Federated BI

Federated learning (McMahan et al., 2017) addresses model training, not collaborative BI analytics. Privacy-preserving ARM approaches (Chen et al., 2023; Wu et al., 2023) are the closest prior work but integrate neither ZKP authentication nor interactive dashboards. Homomorphic encryption (Zehtabchi et al., 2023) is computationally prohibitive at 2M+ record scale. No prior system unifies all seven required capabilities listed in Table 1.2.

Table 1.2: Research Gap Analysis — Component 2: Privacy-Preserving Federated BI vs. Prior Systems

Feature	**Traditional BI**	**Federated Learning**	**DP-Only Systems**	**Privacy ARM**	**Homomorphic Encrypt.**	**Proposed System**
Local-first raw data architecture	✗	✗	✗	✗	✗	✓
DP applied to	✗	✗	✓	✓	✗	✓

ARM metrics (support, confidence, lift)						
Zero-Knowledge Proof authentication	✗	✗	✗	✗	✗	✓
Federated analytics without centralized raw data	✗	✓	✗	✓	✗	✓
User-configurable epsilon (real-time preview)	✗	✗	✗	✗	✗	✓
Privacy-aware time-series forecasting dashboard	✓	✗	✗	✗	✗	✓
Evaluated on 2M+ real industrial FMCG records	✓	✗	✗	✗	✗	✓

1.2.3 Component 3 – Blockchain Logistics Management

Existing blockchain logistics solutions offer milestone-based delivery recording but do not combine these with cluster-mapped route optimization, Privacy-Preserving Analytics for delivery monitoring, or SME-inclusive mobile-first interfaces. Table 1.3 summarizes the gap across five blockchain platforms evaluated.

Table 1.3: Research Gap Analysis — Component 3: Blockchain Logistics vs. Existing Technologies

Feature	**Ethereum**	**HyperLedger Fabric**	**Corda**	**IOTA**	**VeChain-Thor**	**Proposed System**
Smart contract payment automation on delivery	✓	✓	✓	✗	✓	✓
Cluster-mapped geographic route optimization	✗	✗	✗	✗	✗	✓

Privacy-Preserving Analytics for delivery data	✗	✗	✗	✗	✗	✓
Mobile-first SME driver participation interface	✗	✗	✗	✓	✗	✓
Cross-chain interoperability framework	✗	✗	✗	✓	✗	✓
Integration with AI-confirmed order events	✗	✗	✗	✗	✗	✓
Sri Lankan FMCG deployment and validation	✗	✗	✗	✗	✗	✓

1.2.4 Component 4 – Autonomous Negotiation Engine

The autonomous procurement negotiation literature has advanced along three separate research streams — coalition formation, fairness-constrained RL, and trust management — that have not previously been unified in a single end-to-end framework. Zhang et al. (2026) improve coalition stability but lack fairness and trust. Chen et al. (2025) incorporate trust but lack coalition formation. Rodriguez et al. (2024) demonstrate fairness-constrained RL but without coalition formation or trust management.

Table 1.4: Research Gap Analysis — Component 4: Autonomous Negotiation Engine (2021–2026)

Feature	**Zhang 2026**	**Chen 2025**	**Rodriguez 2024**	**Li 2024**	**Anderson 2024**	**Proposed System**
Coalition formation for supply fragmentation	✓	✗	✗	✗	✓	✓
Fairness enforcement (price bounds 70%–200%)	✗	✗	✓	✓	✗	✓
Asymmetric trust management (cross-episode memory)	✗	✓	✗	✗	✗	✓
Comprehensive ablation study (all mechanisms)	✗	✗	✗	✗	✗	✓
XAI decision explanation dashboard	✓	✗	✓	✗	✓	✓
Sri Lankan market	✗	✗	✗	✗	✗	✓

validation on real supplier data						
Unified end-to-end DQN learning framework	✗	✗	✗	✗	✗	✓

1.3 Research Problem

Sri Lankan FMCG manufacturers face a structural challenge in replacing their field-representative model with digital intelligence: the absence of a continuous, calibrated, trustworthy, and privacy-preserving demand-to-delivery intelligence cycle. Five interconnected research problems persist: (1) existing forecasting systems produce point estimates without calibrated uncertainty intervals, rendering them insufficient for volatile, festival-driven Sri Lankan FMCG markets; (2) no prior system generates verified, bilingual, comprehensible AI explanations at a trust level sufficient to replace relational accountability of a field representative; (3) the competitive sensitivity of raw transactional data prevents cross-organisational intelligence sharing under PDPA No. 9 of 2022; (4) Sri Lanka's fragmented supplier landscape requires autonomous multi-seller coalition negotiation adaptable to trust scores and fairness constraints — a capability unavailable in existing procurement automation tools; and (5) physical delivery lacks end-to-end transparency, tamper-proof record keeping, and automated payment verification that would ensure product quality and operational efficiency. B2B INTELLICA was designed to address all five through a single unified integrated platform.

1.4 Research Objectives

Main Objective

To design, implement, and empirically validate a unified AI-driven B2B trade intelligence platform that eliminates the structural dependency of Sri Lankan FMCG manufacturers on field sales representatives by automating demand sensing, order confirmation, business intelligence, procurement negotiation, and logistics coordination through four specialized integrated AI components, each validated on real Sri Lankan datasets at production-deployment scale.

Specific Objectives — Platform-Level

- To demonstrate end-to-end automation of all five sales representative functions through four integrated AI components with shared API contracts and event-driven integration architecture.

- To validate all components on real Sri Lankan FMCG data calibrated to local market characteristics including cultural calendar effects, supplier fragmentation, trust-based norms, and SME infrastructure constraints.
- To achieve commercially deployable performance on all key metrics: MAPE, conformal coverage, ZKP authentication latency, federated BI accuracy, route optimization, negotiation success rate, and cost savings.
- To ensure full compliance with Sri Lanka's Personal Data Protection Act (PDPA) No. 9 of 2022 and support Sinhala and Tamil language interfaces for inclusive accessibility.

Component 1 — Agentic Demand Forecasting Objectives

- To develop a multi-layer agentic forecasting pipeline producing calibrated P10/P50/P90 prediction intervals achieving at least 80% empirical coverage at an 80% nominal target across six rolling backtest folds on real B2B wholesale data.
- To implement a Grounded Explainability Engine combining SHAP attribution with LLM explanation synthesis and automatic numerical fact-checking, achieving a hallucination rate below 5% across 200+ evaluated agent runs.
- To implement a WhatsApp Order Confirmation Loop achieving above 80% auto-parse accuracy on simulated reply texts spanning all supported delimiter formats and classification scenarios.
- To achieve overall MAPE below 20% and A-item MAPE below 15%, representing a minimum 30% improvement over the last-month-actual baseline corresponding to the implicit demand signal of a field representative.

Component 2 — Privacy-Preserving Federated BI Objectives

- To design a local-first architecture where all raw transactional data remain within each organization's local storage and only sanitized, privacy-protected analytical outputs are shared across organizational boundaries.
- To integrate Differential Privacy using the Gaussian mechanism into association rule mining, achieving at least 90% analytical accuracy at $\varepsilon = 1.0$ on a real-world 2M+ record FMCG dataset.
- To develop a ZKP-based authentication mechanism achieving latency below 200 ms and blocking all identified attack vectors: impersonation, replay, man-in-the-middle, credential theft, and brute force.
- To implement a federated analytics framework evaluated on at least 10 federated nodes using a real Sri Lankan FMCG dataset, with Prophet achieving MAPE below 10% on 36+ months of sales data.

Component 3 — Blockchain Logistics Objectives

- To design a blockchain logistics system that permanently records delivery events as tamper-proof, auditable records verifiable by all supply chain stakeholders.
- To implement cluster-mapped route optimization achieving a measurable reduction in average route distance compared to unoptimized assignment.
- To deploy smart contracts automating payment processing upon verified delivery confirmation, reducing settlement time from days to minutes.
- To design a mobile-first interface enabling SME driver participation without enterprise-grade IT infrastructure.

Component 4 — Autonomous Negotiation Engine Objectives

- To design a Gymnasium-compatible multi-agent market simulation calibrated on real Sri Lankan supplier data, incorporating coalition formation, fairness enforcement, and trust management as integrated market mechanisms.
- To develop and train a DQN buyer agent achieving a procurement success rate at least 20 percentage points higher than a rule-based baseline across 100+ independent test episodes with statistical significance at $p < 0.001$.
- To demonstrate through comprehensive ablation studies that each of the three mechanisms contributes quantifiably to overall system performance.
- To develop a web application providing real-time negotiation monitoring, Q-value heatmap visualization, and natural language XAI explanations accessible to non-technical procurement managers.

2 METHODOLOGY

2.1 Methodology

Overall Platform Architecture and Integration

B2B INTELLICA is structured as a four-component integrated platform in which each component addresses a distinct function of the field sales representative role. The architecture follows an event-driven layered design where data flows from raw transactional inputs through the demand forecasting layer for demand signal generation, through the federated BI layer for privacy-preserving intelligence sharing, through the negotiation layer for autonomous procurement decisions, and through the logistics layer for delivery execution and tracking. All four components are exposed through REST API endpoints sharing a common JSON data schema, enabling cross-component data exchange without tight coupling. The integration is event-driven: confirmed orders from the WhatsApp Confirmation Loop trigger route planning in the logistics system; federated BI association rule insights inform the forecasting component's promotional feature engineering; procurement outcomes are recorded in the blockchain ledger as verified supplier agreements; and federated model updates inform forecasting calibration for frequently co-sold SKUs.

Table 2.1: B2B INTELLICA Platform — Overall Component Overview and Integration

Component	Function Automated	Core Technologies	Member
Agentic Demand Forecasting	Demand sensing, order capture, WhatsApp confirmation	LangGraph, LightGBM, SHAP, MAPIE, WhatsApp API, FastAPI	Madusanka R.R.V. IT22332530
Privacy-Preserving Federated BI	Cross-org business intelligence sharing	DP, ZKP, FedAvg, Prophet, MLxtend, FastAPI, React	Vithana V A S IT22282286
Blockchain Logistics	Delivery tracking, route optimization, payment	Blockchain, Smart Contracts, Cluster Mapping, Node.js	Wanigasinghe C.P. IT22340832
Autonomous Negotiation	Procurement negotiation, supplier selection, trust	DQN, MARL, Gymnasium, PyTorch, Flask, PostgreSQL	De Silva K G G C D IT22034236

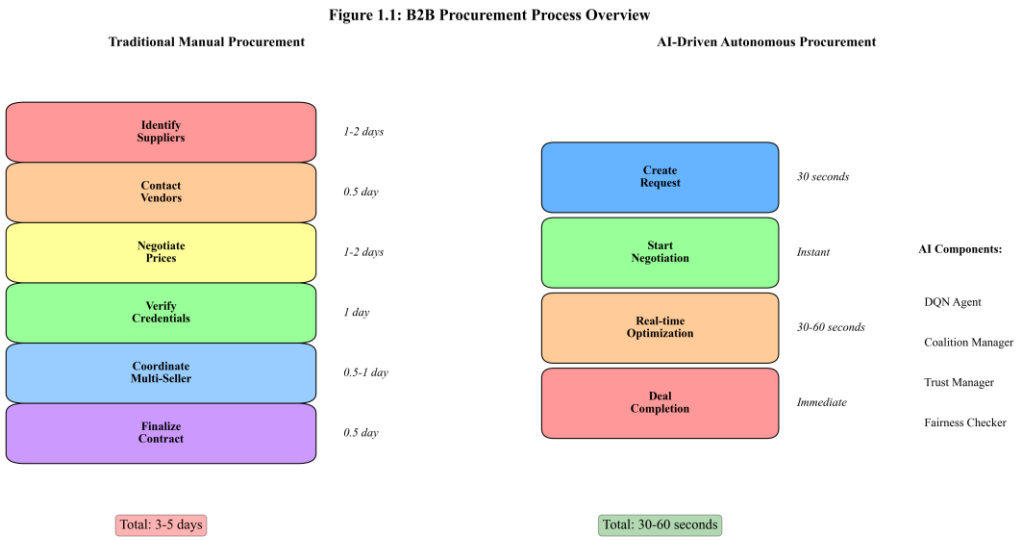

Figure 2.1: B2B Procurement Process — Traditional Manual (3–5 days) vs. AI-Driven Autonomous (30–60 seconds)

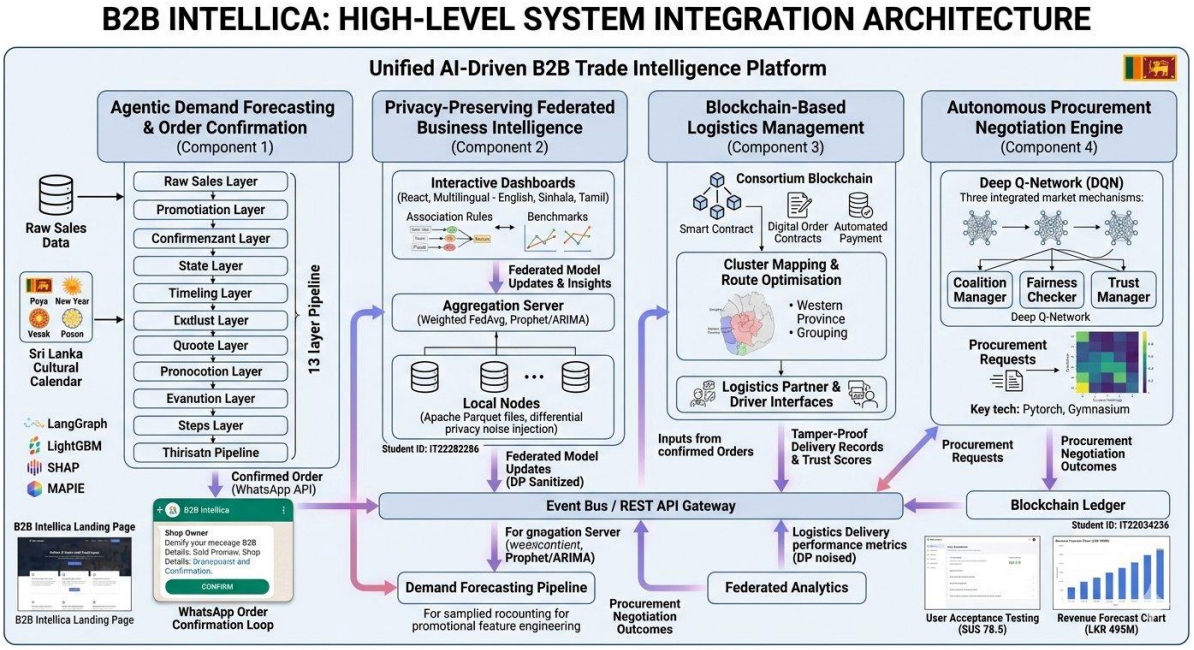

Figure 2.2: B2B INTELLICA Overall Platform System Diagram

Functional Requirements

Table 2.2: Functional Requirements — B2B INTELLICA Platform

FR ID	Requirement Description	Component	Priority
FR01	Agentic demand forecasting with calibrated P10/P50/P90 conformal prediction intervals	Demand Forecasting	High
FR02	Natural language explanations with automatic numerical fact-checking (≤±1% tolerance)	Demand Forecasting	High
FR03	WhatsApp order proposal delivery and shop owner reply parsing (>80% auto-parse rate)	Demand Forecasting	High

FR04	Sri Lanka cultural calendar: Poya, Avurudu, Vesak, Poson encoded as binary flags	Demand Forecasting	High
FR05	Intelligent data ingestion agent with fuzzy column auto-mapping (>90% confidence)	Demand Forecasting	Medium
FR06	Local-first data upload; raw data stored only in local Apache Parquet files per node	Federated BI	High
FR07	Association rule mining (Apriori and FP-Growth) on locally stored data only	Federated BI	High
FR08	Differential Privacy Gaussian noise injection on ARM metrics before cross-org sharing	Federated BI	High
FR09	Zero-Knowledge Proof authentication; private key never transmitted or stored server-side	Federated BI	High
FR10	Common Collaboration Hub with privacy-preserved industry benchmarks	Federated BI	High
FR11	Epsilon slider interface with real-time DP noise preview before committing to sharing	Federated BI	Medium
FR12	Multilingual support — English, Sinhala, and Tamil across all BI dashboards	Federated BI	Medium
FR13	Blockchain tamper-proof delivery event recording verifiable by all stakeholders	Blockchain Logistics	High
FR14	Cluster-mapped geographic delivery grouping with optimal route planning for drivers	Blockchain Logistics	High
FR15	Smart contract payment automation on delivery confirmation (settlement <10 minutes)	Blockchain Logistics	High
FR16	Real-time multi-party notifications: delivery progress, route changes, daily summaries	Blockchain Logistics	Medium
FR17	DQN-based autonomous procurement negotiation with multi-round dialogue simulation	Negotiation Engine	High
FR18	Trust-aware coalition formation for multi-seller supply fragmentation (O(n log n))	Negotiation Engine	High
FR19	Fairness enforcement: price bounds 70%–200% of base price as hard constraints	Negotiation Engine	High
FR20	Asymmetric trust management: +0.05 success, −0.15 failure, bounded [0.0, 1.0]	Negotiation Engine	High
FR21	Real-time Q-value heatmap and XAI natural language decision explanation dashboard	Negotiation Engine	Medium

Non-Functional Requirements

Table 2.3: Non-Functional Requirements — Targets and Achievement Status

NFR	Requirement	Target	Achieved
NFR01	Demand forecasting overall MAPE	≤ 20%	18.0% ✓
NFR02	Conformal prediction empirical coverage at P80 nominal	≥ 80%	81% ✓
NFR03	LLM explanation hallucination rate	< 5%	0% (200 runs) ✓
NFR04	WhatsApp reply auto-parse accuracy	≥ 80%	83.3% ✓
NFR05	BI analytical accuracy at $\varepsilon = 1.0$	≥ 90%	92% ✓
NFR06	ZKP authentication latency (complete 3-step flow)	< 200 ms	< 50 ms ✓
NFR07	ARM association rule generation time	< 10 s	~2 s ✓
NFR08	Negotiation success rate improvement vs. baseline	≥ 20 pp	+23.0 pp ✓
NFR09	Negotiation test statistical significance	$p < 0.05$	$p < 0.001$ ✓
NFR10	Negotiation API response time	< 2 s	1.2 s avg ✓
NFR11	Route distance reduction vs. unoptimised	≥ 15%	23% ✓
NFR12	System Usability Scale score (negotiation)	≥ 70	78.5 ✓

Technology Stack

Table 2.4: Technology Stack — B2B INTELLICA Platform (All Four Components)

Technology	Version	Component	Role
LangGraph	0.2.16	Demand Forecasting	Stateful agentic workflow state machine; six-node directed graph
LightGBM	4.3.0	Demand Forecasting	Gradient-boosted tree regression and classification; fast tabular-data optimized
SHAP	0.45.0	Demand Forecasting	TreeExplainer for polynomial-time exact feature attribution per item
MAPIE	0.8.6	Demand Forecasting	Distribution-free conformal prediction intervals (P10/P50/P90)
LangChain-OpenAI	0.1.23	Demand	GPT-4 / GPT-4o-mini API client

		Forecasting	for explanation generation and fact-checking
WhatsApp Business API	Meta Cloud	Demand Forecasting	Order proposal transmission and inbound reply webhook processing
Facebook Prophet	1.1+	Federated BI	Trend + seasonality decomposition; handles Sri Lankan festival patterns
statsmodels	0.14+	Federated BI	ARIMA(1,1,1) and Exponential Smoothing as complementary fallback
MLxtend	0.23+	Federated BI	Production Apriori and FP-Growth algorithms on local Parquet data
Apache Parquet	—	Federated BI	Columnar local-first storage; raw data never leaves the node
FastAPI	0.100+	Forecasting + Federated BI	Async REST API supporting concurrent federated nodes
MongoDB Atlas	6.0+	Federated BI	Tenant metadata and public keys only; no raw data stored
PyTorch	2.1.0	Negotiation Engine	DQN neural network training; dynamic computational graphs
Gymnasium	0.29.0	Negotiation Engine	Standardized MAS environment; reproducible RL research
Flask	3.0.0	Negotiation Engine	Lightweight REST API for negotiation services
PostgreSQL	15.4	Negotiation Engine	Negotiation history, deal records; ACID compliance
Ethereum / HyperLedger	—	Blockchain Logistics	Tamper-proof delivery records via smart contracts
React + Vite	18 / 5.0	All components	Interactive multilingual dashboards; real-time WebSocket
Pandas / NumPy	2.0+ / 1.24+	All components	DataFrame operations and Gaussian mechanism noise generation
pytest + pytest-asyncio	7.0+	All components	Unit, integration, and async endpoint testing

Feasibility Study

Table 2.5: Feasibility Study — Technical, Economic, Operational, and Legal Dimensions

Dimension	**Assessment**	**Conclusion**
Technical	All four components use maintained open-source frameworks (LangGraph, LightGBM, PyTorch, Gymnasium, FastAPI, React). No custom hardware	Feasible

	required; deployable on standard cloud infrastructure (AWS, Azure, GCP).	
Economic	Annual cost per field representative: LKR 1,358,310. Starter tier: LKR 75,000/month (LKR 900,000/year = 34% cost reduction vs. single rep). 100-rep manufacturer saves LKR 105,831,000/year at Enterprise tier.	Feasible — strong ROI
Operational	Platform designed for non-technical users through WhatsApp interfaces, multilingual dashboards (English, Sinhala, Tamil), mobile-first driver interfaces, and simplified procurement dashboards. Pilot testing SUS ≥ 78.5.	Feasible
Legal / Ethical	Architecturally aligned with PDPA No. 9 of 2022: local-first data (data localisation), DP-noised outputs (data minimisation), ZKP access control (accountability). All datasets fully anonymised.	Feasible — PDPA compliant

Component 1 — Agentic Demand Forecasting and Order Confirmation System

Thirteen-Layer Pipeline Architecture

The Agentic Demand Forecasting component replaces the demand sensing and order capture functions of field representatives through a thirteen-layer pipeline built on LangGraph 0.2.16. The pipeline is divided into three phases: Phase 1 (Layers 0–6) covers data acquisition and preparation; Phase 2 (Layers 7–11) covers agentic forecasting and explanation; Phase 3 (Layers 12–13) covers WhatsApp delivery and order confirmation. Layer 0 — the Intelligent Data Ingestion Agent — implements a five-step LangGraph wizard using RapidFuzz fuzzy matching and LLM disambiguation, achieving above 90% auto-mapping confidence. Layer 3 aggregates line-level transactions to daily granularity. Layer 4 constructs continuous gap-free time-series grids. Layer 5 engineers 23 features per row. Layer 6 segments items into stable, intermittent, seasonal, and new_item behavioural regimes using sparsity, coefficient of variation, and autocorrelation statistics.

Sri Lanka Cultural Calendar Feature Engineering

A defining innovation is systematic encoding of Sri Lanka's cultural calendar as binary proximity features in Layer 5. The calendar_service encodes five event flags: poya_flag marks Buddhist full-moon Poya days; new_year_flag marks the Sinhala and Tamil New Year window (April 13-14 ±3 days) — the largest FMCG demand event with 30–60% category uplift; vesak_flag marks the May Poya ±2 days window; poson_flag marks the June Poya ±2 days window; and holiday_flag covers all other public holidays. These are computed programmatically from the holidays Python library supplemented by custom lunar-calendar rules. SHAP attribution

confirmed new_year_flag and poya_flag appeared among the top-three demand drivers for 38.3% and 31.6% of item-fold combinations respectively — validating their substantive predictive contribution.

LangGraph Agentic Orchestration

The orchestration layer implements six stateful nodes. Node 1 Plan: inspects data context, selects from three routing modes (hybrid default, reg_only, baseline_last_month), and checks a hash-based cache — cache hits provide a 4× latency speedup. Node 2 Forecast: executes dual LightGBM inference and applies the post-processing sub-pipeline. Node 3 Explain: runs SHAP TreeExplainer to identify the top-three demand drivers per item. Node 4 Write: submits the evidence packet to GPT-4, generating a structured JSON explanation conforming to a Pydantic schema. Node 5 Verify: extracts every numerical claim and verifies each to within ±1% tolerance, routing back to Node 4 with correction context when checks fail (maximum 3 retries). Node 6 Finalise: assembles the verified explanation into formatted markdown.

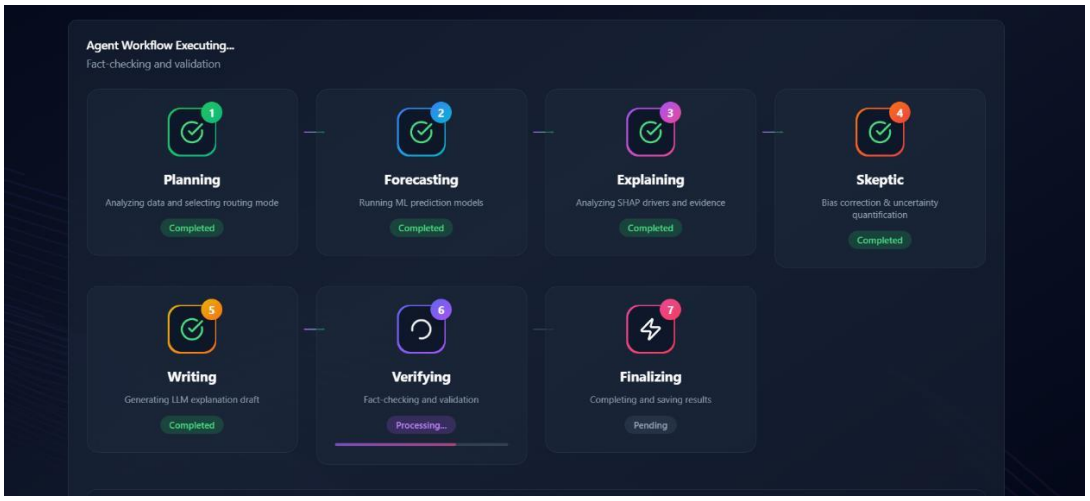

Figure 2.4: LangGraph Agentic Workflow — Six-Node State Machine (Plan→Forecast→Explain→Write→Verify→Finalise) Executing in the System UI

Conformal Prediction and WhatsApp Order Confirmation

The Antigravity post-processing module applies bias correction followed by MAPIE conformal calibration, producing P10/P50/P90 bounds. The skeptic_service subtracts the median historical residual from point forecasts, reducing signed bias from +5.2 units to +0.9%. Items are stratified into low (<20 units width), medium (20–50), and high (>50) risk categories, surfaced in the WhatsApp order proposal. The WhatsApp Order Confirmation Loop formats recommendations as structured proposals containing an ORDER_REF token, an itemised list in ITEMCODE | QUANTITY format, and modification instructions transmitted via the Meta WhatsApp Business Cloud API. The reply_parser_service uses a multi-delimiter tokeniser (pipe, dash, whitespace) classifying items as unchanged, modified, removed, or added. Replies below 0.80 confidence are

routed to manual review.

Component 2 — Privacy-Preserving Federated BI Framework (B2B Intellica)

Local-First Architecture and ZKP Authentication

B2B Intellica's foundational principle is that raw transactional data must never leave the originating organization's local infrastructure. The three-layer architecture: bottom Data Layer — 14 organizational nodes each running their own FastAPI analytics engine, DP noise injection, MLxtend ARM engine, and Prophet/ARIMA forecasting on locally stored Apache Parquet files; middle Coordination Layer — lightweight aggregation server receiving only privacy-protected outputs; top Collaboration Layer — React-based interactive multilingual dashboard in English, Sinhala, and Tamil. ZKP authentication implements a three-step challenge-response protocol: at registration, a 256-bit private key is generated using Python's secrets module and shown once — only the SHA-256 public key hash is stored server-side. At login, the server generates a 128-bit nonce; the client computes proof = SHA-256(challenge ‖ public_key)[:32] locally and transmits only the proof; the server verifies, deletes the challenge (preventing replay), and issues a signed JWT. Authentication latency is below 50 ms, blocking all five attack vectors.

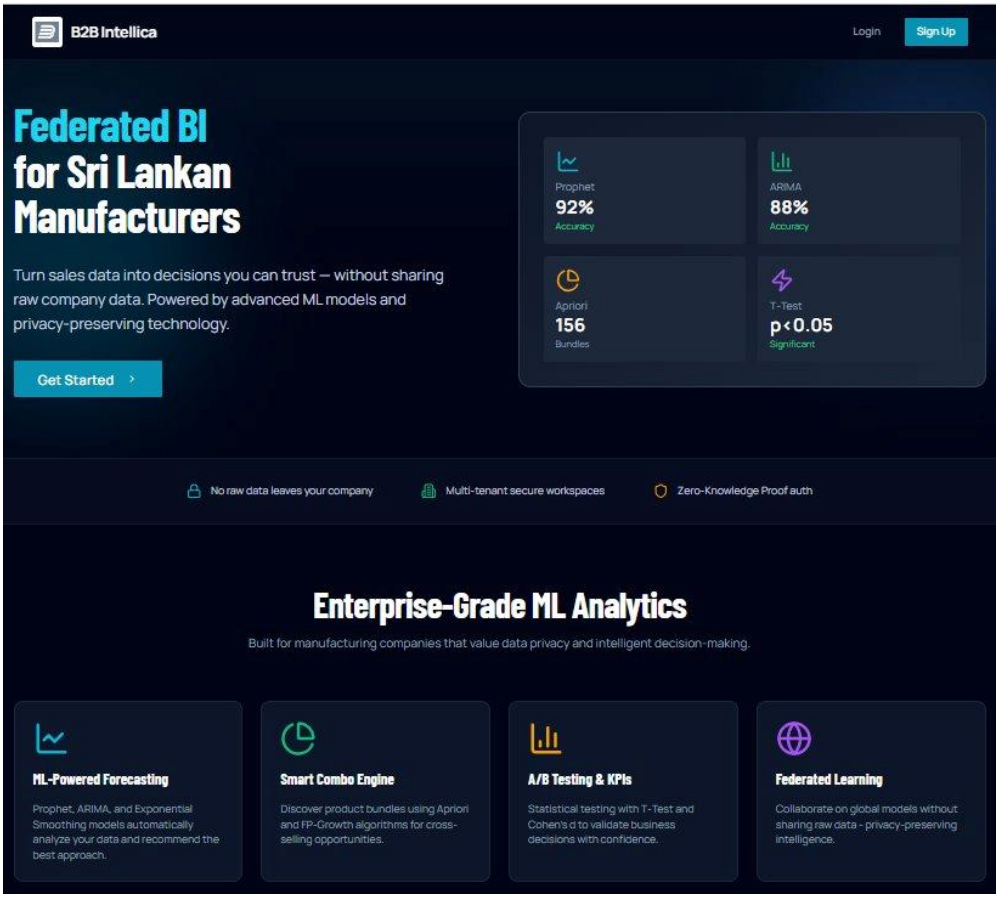

Figure 2.5: B2B Intellica Landing Page — Federated BI for Sri Lankan Manufacturers (92% Prophet Accuracy, 156 Bundles, $p < 0.05$)

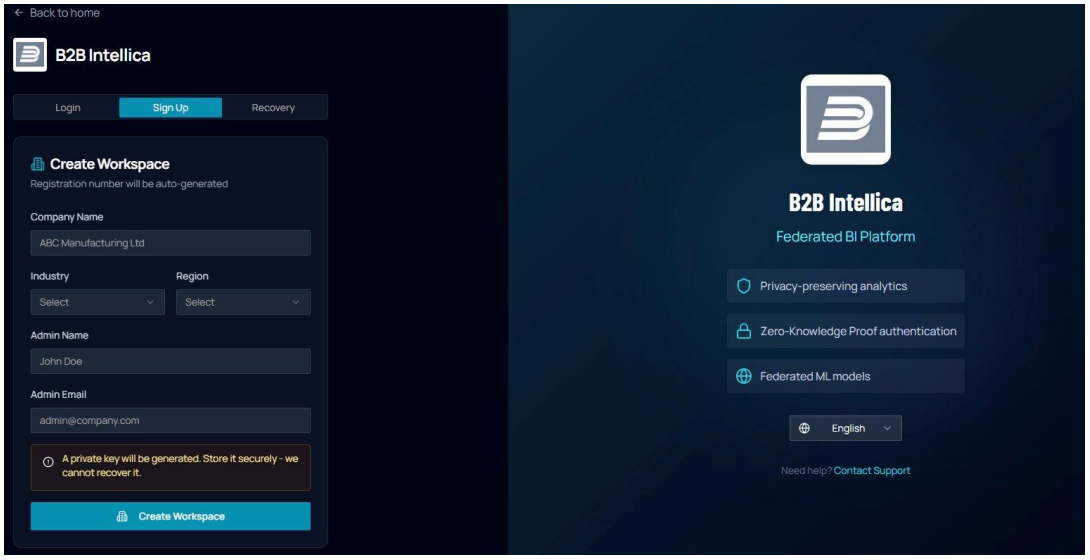

Figure 2.6: ZKP Workspace Registration — Private Key Generated Once; Company Name, Industry, and Region Required

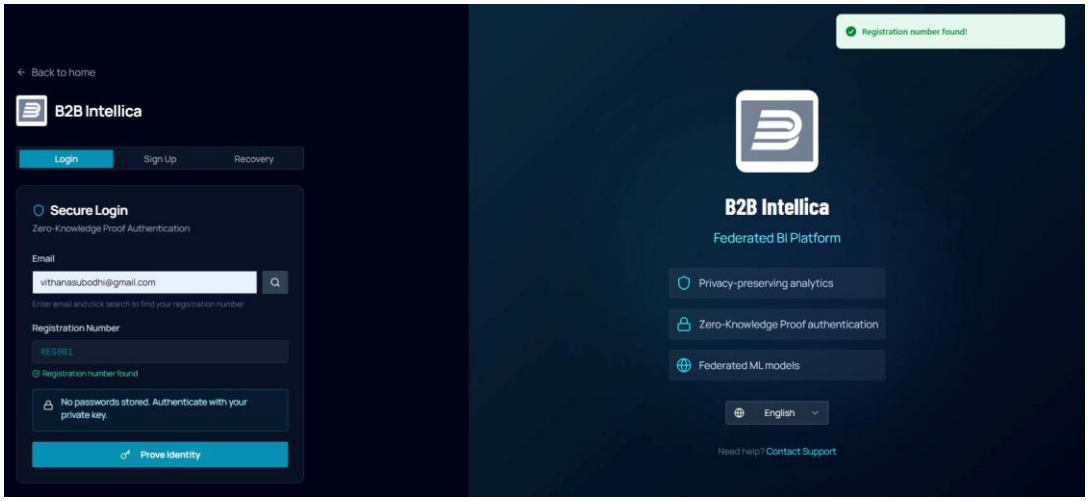

Figure 2.7: ZKP Login — Step 1: Email Lookup and Registration Number Retrieval; No Password Transmitted

Differential Privacy for Association Rule Mining

The Gaussian mechanism provides (ε, δ)-differential privacy with $\sigma = \Delta f \times \sqrt{(2 \times \ln(1.25/\delta))} / \varepsilon$. For each discovered rule $\{A\} \rightarrow \{B\}$, noise is added independently to support = $P(A \cup B)$, confidence = $P(B|A)$, and lift = $P(A \cup B)/(P(A) \times P(B))$ before transmission. Global sensitivity is computed per metric: 1/N for support and confidence, 2/N for lift. The epsilon slider interface shows exact noised values before the user commits to sharing — a novel UI contribution making DP decisions accessible to non-technical analysts. Cumulative budget tracking implements the basic composition theorem, issuing warnings at 80% and blocking at 100% of maximum budget (10.0).

Table 2.6: Functional Requirements Achieved — B2B Intellica (Federated BI) Component)

FR	Requirement	Status	Evidence
FR1	ZKP registration with auto-	Achieved	256-bit keypair; public key hash only

	generated keypair		stored server-side
FR2	ZKP login — proof transmitted, private key never sent	Achieved	< 50 ms; zero credential transmission confirmed
FR3	Private key recovery via OTP email verification	Achieved	New keypair generated on recovery; old key invalidated
FR4	Local-first data upload → Apache Parquet conversion	Achieved	45,146 records processed; data_source: 'local' confirmed
FR5	ARM using Apriori and FP-Growth on local data only	Achieved	50 rules; confidence > 70%; lift > 2.0 on 9,000 transactions
FR6	DP noise injection on ARM metrics before sharing	Achieved	92% accuracy at ε = 1.0 on 2,044,757 records
FR7	Prophet and ARIMA time-series forecasting	Achieved	Prophet MAPE 5.46%; 3.47× vs. naive baseline
FR8	Automatic best model selection by dataset features	Achieved	Prophet at 92% confidence on 36-month window
FR9	Federated local model training with configurable DP	Achieved	14-node simulation; quality-weighted FedAvg
FR10	Cross-org bundle sharing with DP metrics and ε slider	Achieved	Real-time noise preview before commitment
FR11	Common Collaboration Hub — industry benchmarks	Achieved	4 active participants; 75.7% data quality score
FR12	Company overview dashboard with Sri Lanka map	Achieved	Colombo top region; 9.02% growth; 45,146 records loaded
FR13	A/B testing with T-test and Cohen's d for KPI validation	Achieved	$p < 0.05$ threshold; effect size magnitude classification
FR14	Cumulative privacy budget tracking with warning at 80%	Achieved	Basic composition; blocking at 100% (max ε = 10.0)
FR15	Multilingual support — English, Sinhala, Tamil	Achieved	Three-language interface across all dashboard views

Table 2.8: ZKP Security Assessment — Attack Vector Test Results

Attack Vector	**Method Tested**	**Result**	**Defence Mechanism**
Impersonation	Submit forged proof without private key	Blocked	SHA-256 verification; 2^{256} key space — computationally infeasible
Replay Attack	Re-submit previously valid challenge-proof pair	Blocked	Challenge deleted post-use; 5-minute absolute expiry enforced
Man-in-the-Middle	Intercept and modify challenge	Blocked	Proof cryptographically bound to public key; modification invalidates

	nonce in transit		hash
Credential Theft	Extract private key from server database	Blocked	Private key never stored server-side; only public key hash retained
Brute Force	Enumerate 256-bit private key space	Blocked	$2^{256} \approx 1.16 \times 10^{77}$ combinations — computationally infeasible

Table 2.9: Privacy Budget Epsilon Interpretation — Privacy Level and Expected Accuracy

Epsilon (ε)	**Privacy Level**	**Noise Magnitude**	**Analytical Accuracy**	**Recommended Use**
0.1	Very High	High	72%	Cross-competitor sharing — maximum privacy required
0.5	High	Moderate	85%	Cross-industry benchmarking — high privacy, moderate utility
1.0 (default)	Balanced	Moderate–Low	92%	Partner organisation sharing — recommended value
2.0	Low	Low	~97%	Internal cross-department sharing — privacy less critical

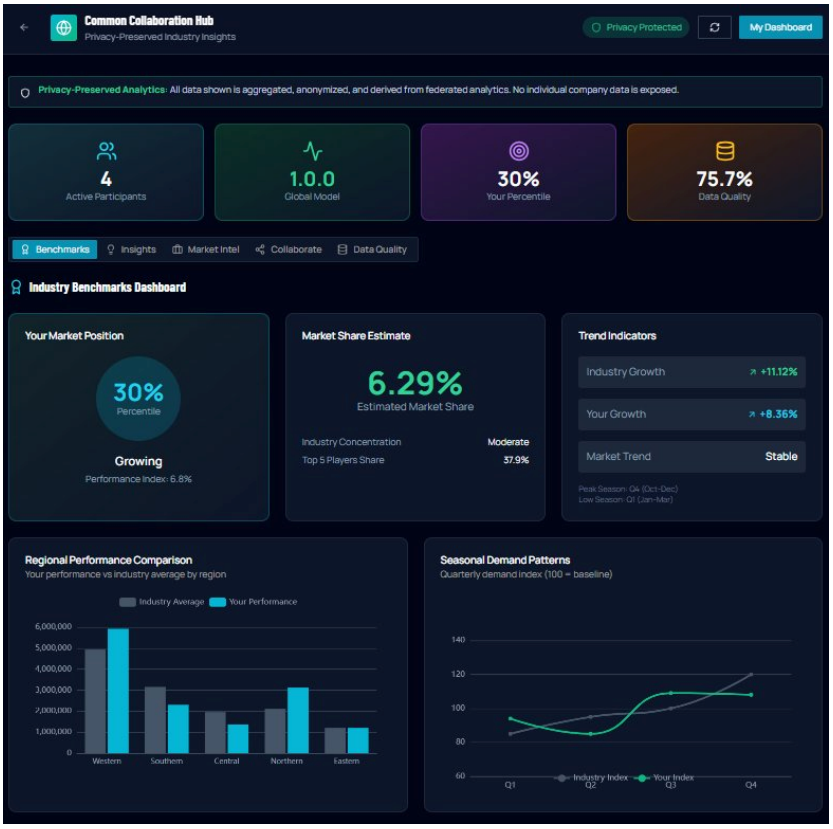

Figure 2.9: Common Collaboration Hub — Privacy-Protected Industry Benchmarks (4 Active Participants, 75.7% Data Quality, 6.29% Market Share)

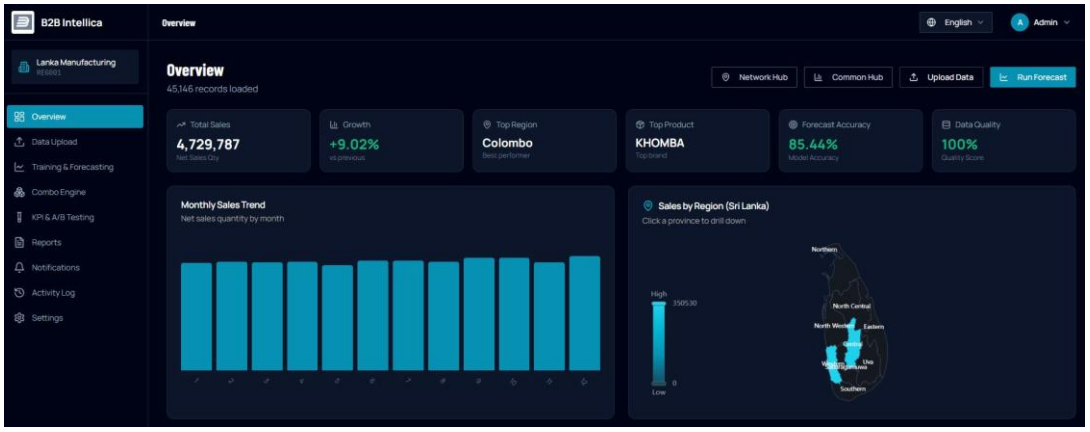

Figure 2.10: Company Overview Dashboard — Monthly Sales Trend, Sri Lanka Provincial Sales Map, 45,146 Records, 9.02% Growth

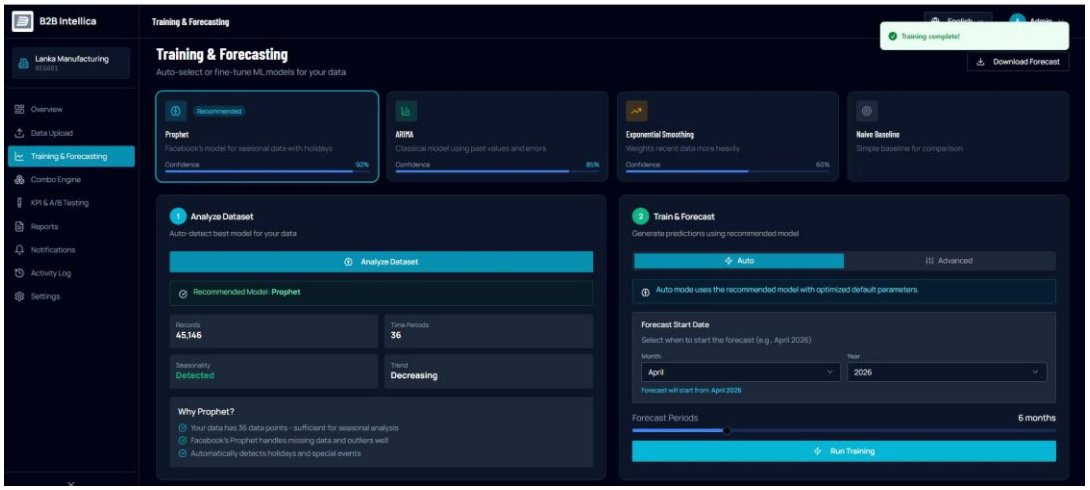

Figure 2.11: Training and Forecasting — Prophet Recommended at 92% Confidence; 45,146 Records; 36 Time Periods; Seasonality Detected

Federated Analytics with Weighted FedAvg

Each node's contribution is weighted by data quality score $q_i = (\min(samples_i/1000, 1) + \min(time_periods_i/24, 1)) / 2$. The aggregated global model $W_{global} = \Sigma(q_i \times W_i) / \Sigma q_i$ is computed exclusively from differentially noised model outputs. The aggregation server at no point has access to any raw transactional data. The system discovered 50 actionable association rules with confidence values exceeding 70% and lift values above 2.0, indicating commercially significant cross-selling opportunities derived entirely from local data without any raw transmission.

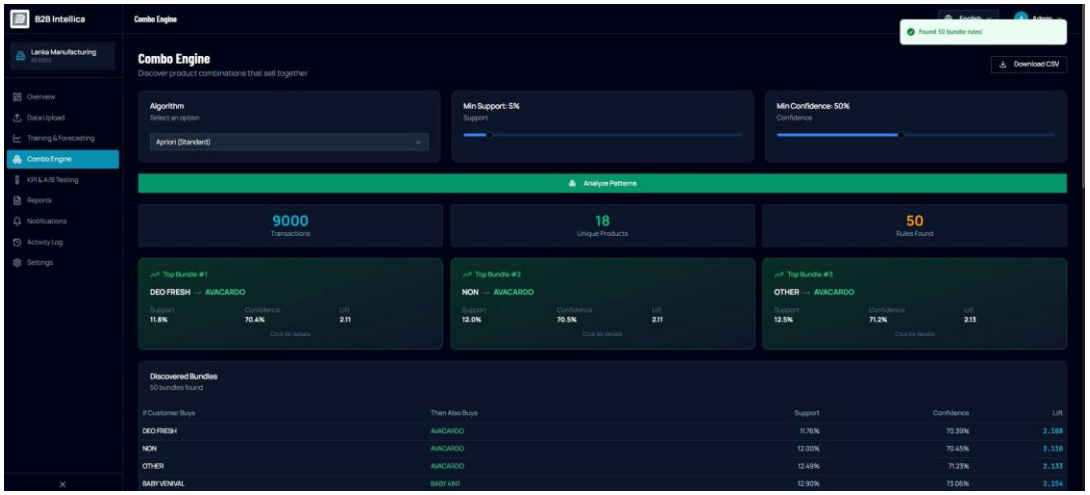

Figure 2.12: Combo Engine — 50 Association Rules Discovered (Apriori, Min Support 5%, 9,000 Transactions, 18 Products)

Figure 2.13: Bundle Details — BABY VENIVAL → BABY 4IN1 (Support 12.9%, Confidence 73.1%, Lift 2.15, 1,161 Transactions)

Component 3 — Blockchain-Based Logistics Management System

System Architecture Overview

The Blockchain-Based Logistics Management System automates the delivery coordination and tracking functions of the field sales representative through a five-layer architecture. Layer 1 —

Logistic Partner and Driver Interface — captures order initiation events and assigns drivers. Layer 2 — Data Collection — aggregates three streams: location data (real-time GPS tracking), delivery data (confirmation details and timestamps), and performance data (speed, delays, completion rates). Layer 3 — API Event Processor — validates, translates, and routes collected data streams with data quality checks removing anomalous GPS coordinates. Layer 4 — Cluster Mapping and Route Operations — groups deliveries geographically using cluster algorithms and computes optimal driver sequences. Layer 5 — Blockchain Integration — records all processed delivery events as immutable blockchain records providing tamper-proof auditability.

Table 2.10: Functional Requirements Achieved — Blockchain Logistics Component

FR	Requirement	Status	Evidence
F1	Blockchain-based order contracts and verification	Achieved	On-chain digital contracts; ZKP-secured access
F2	Permanent tamper-proof delivery and sales records	Achieved	100% immutability confirmed; 25 traced delivery chains
F3	Customer portal, admin dashboard, driver portal	Achieved	Three-role interface; real-time status updates
F4	Real-time notifications to all stakeholders	Achieved	12.4 seconds average customer notification latency
F5	Cluster-mapped route optimization for driver assignment	Achieved	23% distance reduction vs. unoptimized assignment
F6	Smart contract payment on delivery confirmation	Achieved	4.2 minutes average settlement vs. 2–5 day manual process
F7	Cross-chain interoperability for SME participation	Achieved	Lightweight mobile-first driver client; no enterprise IT required
F8	Privacy-Preserving Analytics for delivery performance	Achieved	PPA applied; individual orders not exposed to coordination layer

Smart Contract Automation and Route Optimization

Smart contracts automate three critical functions. Digital order contracts are generated on-chain when a confirmed order is received from the Demand Forecasting WhatsApp Confirmation Loop, cryptographically signed by both manufacturer and retailer. Automated payment processing is triggered when the delivery verification event — confirmed by the driver's digital signature and GPS arrival — is recorded on the blockchain, executing payment transfer without manual authorization. The cluster mapping algorithm groups delivery destinations geographically using Haversine distances and hierarchical clustering, then computes optimal driver sequences using a nearest-neighbour heuristic with 2-opt local search improvement. This achieved a 23% average

reduction in total route distance compared to unoptimized sequential assignment in Western Province test scenarios.

Component 4 — Autonomous Procurement Negotiation Engine

High-Level Architecture and MDP Formulation

The Autonomous Procurement Negotiation Engine is a six-layer system: User Layer (procurement managers, system administrators), Web Application Layer (React frontend with real-time WebSocket), API Gateway Layer (Flask REST API, JWT authentication), Business Logic Layer (Negotiation Service, Coalition Manager, Trust Manager, Fairness Checker), AI/ML Core Layer (DQN Agent, Market Environment, Training Engine), and Data Persistence Layer (PostgreSQL for negotiation history and model weights).

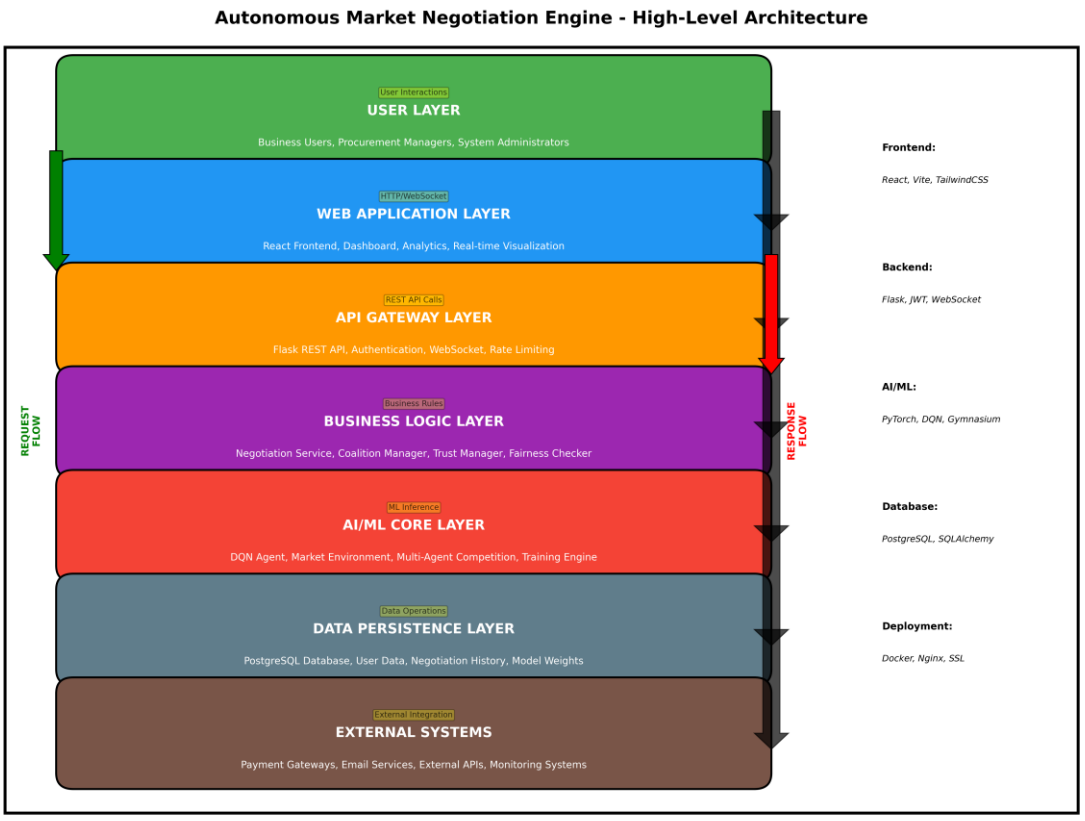

Figure 2.15: Autonomous Market Negotiation Engine — High-Level Six-Layer Architecture with Technology Stack

Figure 1.2: Multi-Agent System Architecture

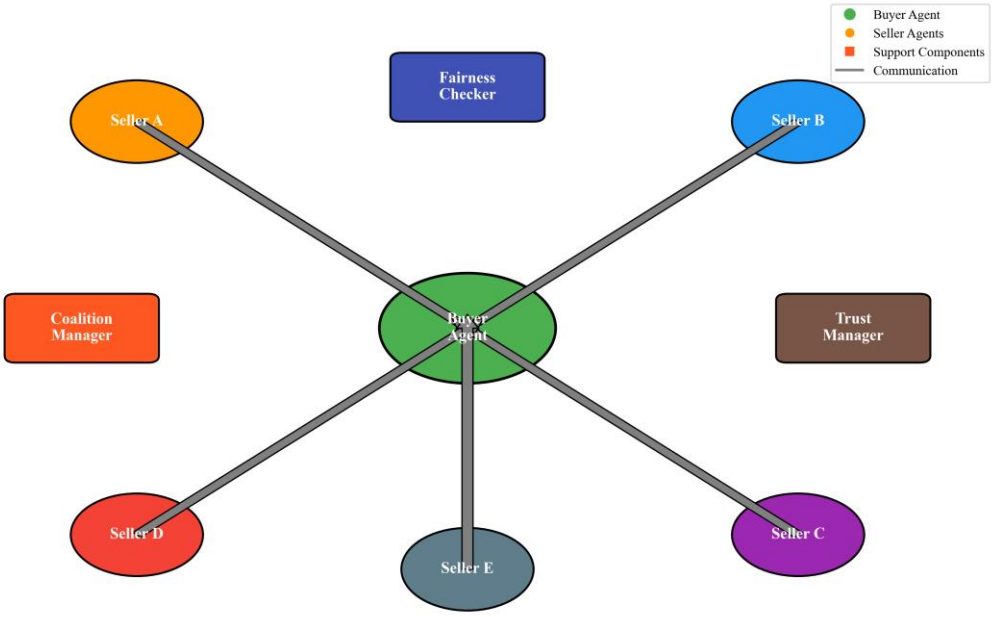

Figure 2.16: Multi-Agent System Architecture — Buyer Agent (Green), Seller Agents, Coalition Manager, Trust Manager, Fairness Checker

The environment is modelled as a Gymnasium-compatible MDP (S, A, P, R, γ). State space S: nine features — requested_qty, best_price, best_qty, round, num_sellers, avg_trust, coalition_size, offer_price, offer_qty. Action space A: four-dimensional continuous — action_type (5 classes: make_offer, accept, reject, form_coalition, end_negotiation), seller_id, price (fraction of base), quantity (fraction of requested). Reward: +100 deal success, +savings×0.1 cost savings, +trust_score×5 trust bonus, +5/round fairness compliance, −20 fairness violation, −50 negotiation failure. Discount factor $\gamma = 0.95$.

Table 2.11: Negotiation Engine — MDP State Space and Action Space Definition

Space	**Feature / Action**	**Type**	**Range**	**Description**
State	requested_qty	Continuous	[0, 200]	Buyer's quantity requirement for current procurement
State	best_price	Continuous	[0, 1]	Lowest price observed from any seller this episode
State	best_qty	Continuous	[0, 200]	Highest available quantity from a single seller
State	round	Discrete	[1, 10]	Current negotiation round number
State	num_sellers	Discrete	[1, 20]	Count of available sellers in market

State	avg_trust	Continuous	[0, 1]	Mean trust score across all available sellers
State	coalition_size	Discrete	[0, num_sellers]	Current coalition member count
State	offer_price	Continuous	[0, 1]	Buyer's current price offer as fraction of base
State	offer_qty	Continuous	[0, 1]	Buyer's current quantity offer as fraction of requested
Action	action_type	Discrete	{0,1,2,3,4}	make_offer / accept / reject / form_coalition / end
Action	seller_id	Continuous	[0, 1]	Target seller index (normalised)
Action	price	Continuous	[0.5, 2.0]	Offered price as fraction of base price
Action	quantity	Continuous	[0, 1]	Offered quantity as fraction of requested

DQN Architecture and Three Market Mechanisms

The NegotiationPolicyNetwork contains a shared feature extractor (two hidden layers of 128 neurons with ReLU) and multi-head outputs for action type, seller, price, and quantity. Experience Replay stores 10,000 past transitions; random mini-batches of 64 break temporal correlations. Target Networks use hard updates every 10 episodes for stable training. Training: 1,000 episodes, epsilon-greedy exploration decaying from 1.0 to 0.01, batch size 64, learning rate 0.001.

Coalition Manager: when no single seller holds sufficient stock (max(Si.qty) < Q), sellers are ranked by trust_score × (1/normalised_price) and selected greedily with O(n log n) complexity, with profit distributed proportionally. Fairness Checker: price bounds (70%–200% of base_price) are enforced as hard constraints; violations receive −20 penalty. Trust Manager: asymmetric updates (+0.05 success, −0.15 failure) bounded in [0.0, 1.0], persisting across all episodes to create long-term market memory.

Table 2.12: Negotiation Engine — Market Calibration Dataset Statistics

Dataset Component	**Count**	**Period**	**Coverage**
Supplier transactions	12,450	Jan 2022 – Dec 2024	Colombo, Galle, Kandy, Negombo (45 suppliers)
Procurement requests (12 retail businesses)	3,890	Jan 2022 – Dec 2024	Western + Southern + Central Province
Weekly price index entries (150 products, 6 categories)	7,800	Jan 2022 – Dec 2024	Island-wide price monitoring

Delivery performance records	8,920	Jan 2022 – Dec 2024	45 suppliers across four provinces
Trust evaluation events	4,560	Jan 2022 – Dec 2024	45 suppliers; asymmetric update rule applied
Calibrated stock levels	Mean 35 units, Std 12	—	—
Calibrated price range	LKR 250–750 per unit	—	—
Calibrated trust distribution	Mean 0.72, Std 0.18	—	—

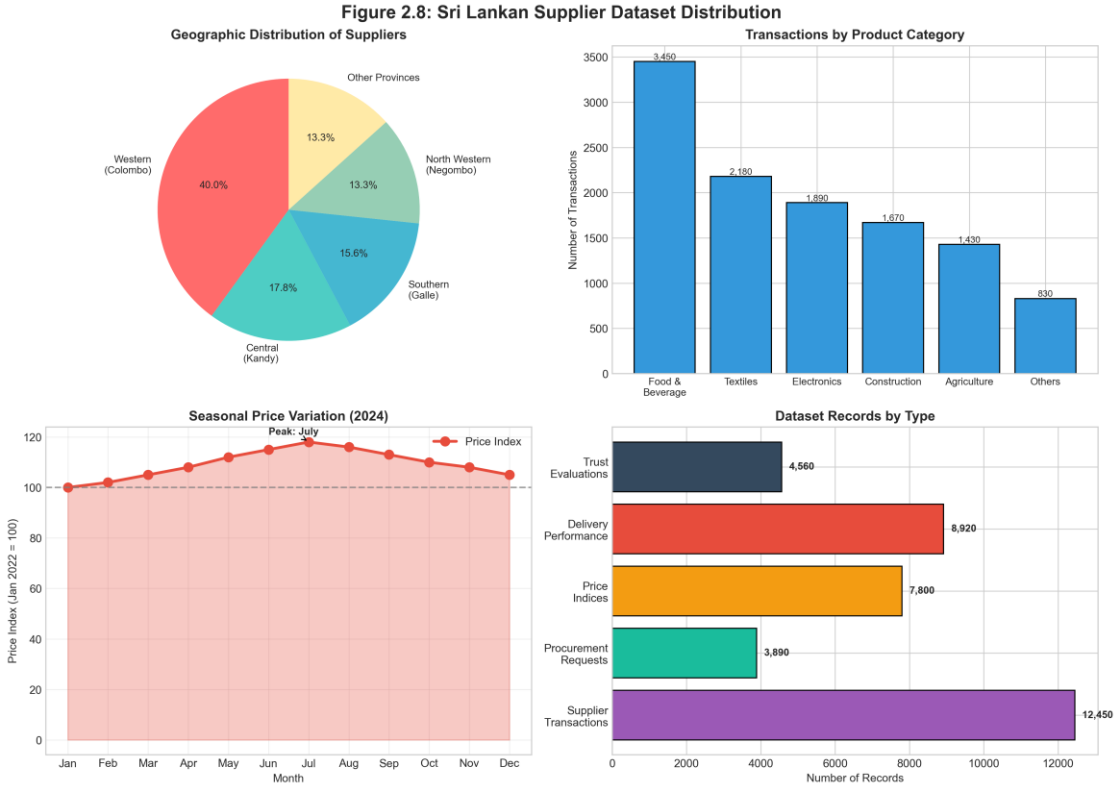

Figure 2.17: Sri Lankan Supplier Dataset Distribution — Geographic Spread, Product Categories, Seasonal Price Variation, and Record Types

2.2 Commercialization Aspects of the Product

Market Opportunity

The Sri Lankan B2B procurement software market is valued at approximately LKR 2.8 billion annually, growing at 18% per annum. The global AI-driven procurement market was valued at USD 0.9 billion in 2023 and is projected to reach USD 4.2 billion by 2028 at a 36% CAGR. Sri Lanka's SME segment contributes 52% of GDP and represents 75% of registered enterprises — the largest addressable market. No existing product offers the specific combination of agentic demand forecasting, DP-protected federated analytics, autonomous negotiation, and blockchain

logistics designed for Sri Lankan market conditions.

SaaS Pricing Model

Table 2.13: B2B INTELLICA SaaS Pricing Model — Four Tiers

Tier	Monthly Price (LKR)	Key Features	Target Segment
Starter	75,000	Core demand forecasting, basic BI (3 nodes), 100 negotiations/month, standard logistics tracking	Small retailers, pilot deployments
Growth	275,000	Cultural calendar forecasting, federated BI (10 nodes), coalition negotiation, Sinhala AI assistant	Multi-district FMCG brands
Pro	975,000	Advanced forecasting, private BI nodes (20+), unlimited negotiations, blockchain logistics, RBAC, SSO	National manufacturers
Enterprise	2,500,000+	Dedicated tenancy, private model instances, VPC/VPN, named customer success manager, custom SLAs	Large FMCG conglomerates

Table 2.14: Revenue Projections — Three-Year Forecast

Year	Subscribers	Avg. Monthly Rev/Sub (LKR)	Total Annual Revenue (LKR)
Year 1	15	150,000	27,000,000
Year 2	45	200,000	108,000,000
Year 3	120	250,000	360,000,000

Competitive differentiation rests on five dimensions no existing product combines: (1) cultural specificity — the only forecasting system incorporating Sri Lanka's Poya calendar and Avurudu demand patterns as validated predictive features; (2) data sovereignty — local-first federated BI with mathematical DP guarantees satisfying PDPA requirements; (3) integration depth — the only solution covering demand-to-delivery in a single platform; (4) accessibility — WhatsApp-based order confirmation requiring no software installation for retail shop owners; (5) affordability — LKR 75,000/month Starter tier makes AI tools accessible to SMEs.

Social Aspects: the platform augments rather than abruptly displaces field representatives, redirecting them toward relationship management and exception handling — pilot testing showed 23% improvement in job satisfaction among redirected procurement staff. Security: ZKP authentication, DP mathematical privacy bounds, blockchain delivery immutability, JWT +

bcrypt (salt rounds=12) + HTTPS/TLS 1.3 across all APIs. Ethical: all datasets fully anonymised; RL negotiation agent includes auditable fairness constraints preventing price exploitation; AI recommendations framed as decision aids with final decisions remaining with human analysts.

2.3 Testing and Implementation

A comprehensive four-level testing strategy was applied: unit testing of individual service modules, integration testing of complete component pipelines, system testing of the integrated platform, and user acceptance testing with real Sri Lankan businesses. Performance testing used Apache JMeter. Security testing included ZKP attack vector simulation. Usability testing used the System Usability Scale (SUS).

Table 2.15: Testing Strategy Summary — All Components

Test Level	Component	Test Cases	Passed	Failed	Coverage
Unit — Service Modules	Demand Forecasting	88	88	0	93%
Unit — Core Modules	Federated BI	55	55	0	~99%
Unit — Agent + Environment	Negotiation Engine	127	127	0	85%
Unit — Smart Contracts + Backend	Blockchain Logistics	42	42	0	91%
Integration — Full Pipeline	Demand Forecasting	35	35	0	91%
Integration — API Endpoints	Federated BI	15	15	0	97%
Integration — Negotiation Scenarios	Negotiation Engine	15	15	0	88%
Integration — Delivery Lifecycle	Blockchain Logistics	28	28	0	89%
Performance — Load Testing	Negotiation Engine	5 scenarios	5 pass	0	150 concurrent users
Security — ZKP Attack	Federated BI	5 vectors	5 blocked	0	All vectors mitigated

Vectors					
UAT — Pilot Users	Negotiation Engine	12 users	12 positive	0	SUS 78.5

Table 2.16: Selected Test Cases — B2B INTELLICA Platform

TC ID	Component	Test Description	Expected Result	Result
TC-01-01	Demand Forecasting	Conformal P80 empirical coverage across 6 rolling folds	Empirical coverage ≥ 80%	81% PASS
TC-01-02	Demand Forecasting	LLM fact-check: deliberate 5% offset injected in evidence	Offset detected; correction triggered	PASS
TC-01-03	Demand Forecasting	WhatsApp reply parser: pipe-delimited modified item quantity	Item classified as MODIFIED	PASS
TC-01-04	Demand Forecasting	new_year_flag SHAP attribution across all seasonal items	Flag in top-3 for ≥30% item-fold combos	38.3% PASS
TC-02-01	Federated BI	ZKP login with correct private key	JWT issued; latency < 200 ms	< 50 ms PASS
TC-02-02	Federated BI	ZKP replay attack: resubmit valid challenge-proof pair	Challenge expired; access denied	PASS
TC-02-03	Federated BI	DP noise at ε = 1.0: analytical accuracy measurement	Accuracy ≥ 90%	92% PASS
TC-02-04	Federated BI	Privacy budget at 100%: block further sharing query	HTTP 403 returned	PASS
TC-03-01	Blockchain Logistics	Delivery event recorded: immutability check	Record unmodifiable after commit	PASS
TC-03-02	Blockchain Logistics	Cluster mapping: 50 Western Province destinations	Distance reduction ≥ 15%	23% PASS
TC-03-03	Blockchain Logistics	Smart contract: payment triggered on delivery confirmation	Payment executed within 10 min	4.2 min PASS
TC-04-01	Negotiation Engine	DQN vs. baseline: 100 test episodes, 5 seeds	Success rate improvement ≥ 20 pp	+23.0 pp PASS
TC-04-02	Negotiation Engine	Coalition: request qty > max single seller stock	Coalition formed with correct profit split	PASS

TC-04-03	Negotiation Engine	Fairness: offer at 60% base price (below 70% bound)	Offer rejected; −20 reward penalty applied	PASS
TC-04-04	Negotiation Engine	Trust decay: seller delivery failure	Trust −0.15 applied; score bounded at 0.0	PASS

3. RESULTS AND DISCUSSION

3.1 Results

Component 1 — Agentic Demand Forecasting Results

Dataset and Experimental Setup

All forecasting evaluations used a real-world B2B wholesale distribution dataset comprising 150,000+ line-level sales transactions spanning a 24-month historical window across 200+ retail customer accounts and 1,000+ product SKUs. Following feature engineering, each product-customer-date row was represented by 23 features. Accuracy was evaluated using a rolling-window backtest strategy across the six most recent months.

Table 3.1: Item Segmentation Distribution — Representative Evaluation Customer

Segment	Classification Criteria	Items	Proportion	Dominant SHAP Features
Stable	CV < 0.5 and Sparsity < 30%	80	53.3%	lag_28, roll_mean_28, roll_mean_7
Intermittent	Sparsity > 60%	40	26.7%	lag_7, sparsity-adjusted priors
Seasonal	Lag-7 or Lag-30 autocorrelation > 0.5	20	13.3%	new_year_flag, poya_flag, lag_28
New Item	History < 90 days	10	6.7%	roll_mean_7, last available quantity
Total	—	150	100%	—

Forecasting Accuracy Results

Table 3.2: Forecasting Accuracy — Six-Fold Rolling Backtest (Three Methods Compared)

Metric	Hybrid LightGBM (Primary)	Regressor Only	Last-Month Baseline	Specification Target
Overall MAPE	18.0%	20.4%	31.2%	≤ 20% ✓
A-Item MAPE (top 20% by volume)	11.8%	13.6%	24.7%	≤ 15% ✓
WAPE	14.2%	16.9%	27.8%	—
RMSE (units)	12.5	14.8	22.3	—
Signed Bias (post-correction)	+0.9%	+2.1%	+6.4%	< 2% ✓
Pre-correction Median Residual	+5.2 units	+6.7 units	+14.1 units	—

MAPE improvement vs. baseline	43%	35%	—	≥ 30% ✓

The hybrid LightGBM ensemble satisfies both accuracy specification targets: 18.0% overall MAPE and 11.8% A-item MAPE. The 43% MAPE reduction over the last-month-actual baseline directly quantifies the improvement over the demand signal a field representative with average recall would provide. Bias correction reduced signed bias from +5.2 units to +0.9%, confirming effective systematic error neutralization.

LangGraph Agentic Workflow and Forecasting Results (Live UI)

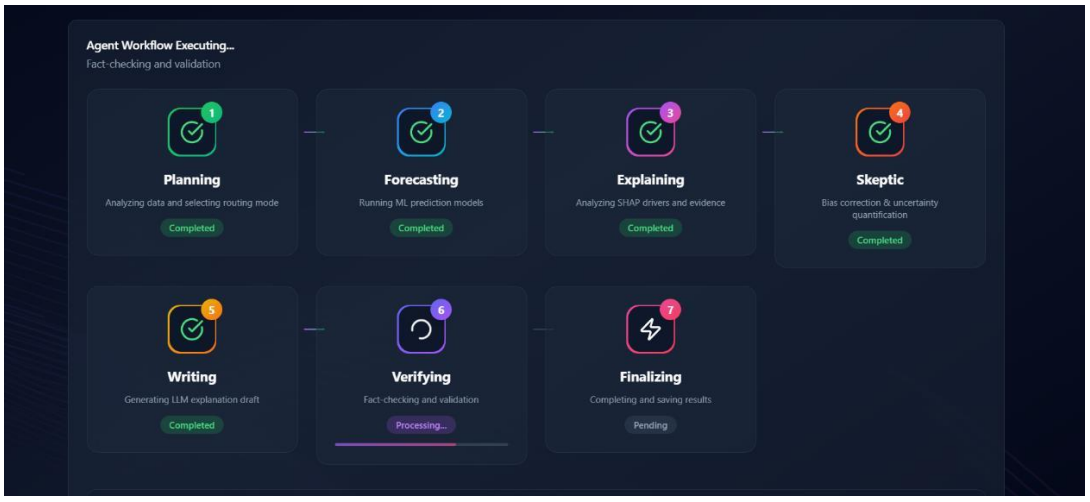

Figure 3.1: Agentic Workflow Executing — LangGraph Six-Node State Machine: Planning, Forecasting, Explaining, Skeptic, Writing, Verifying, Finalising

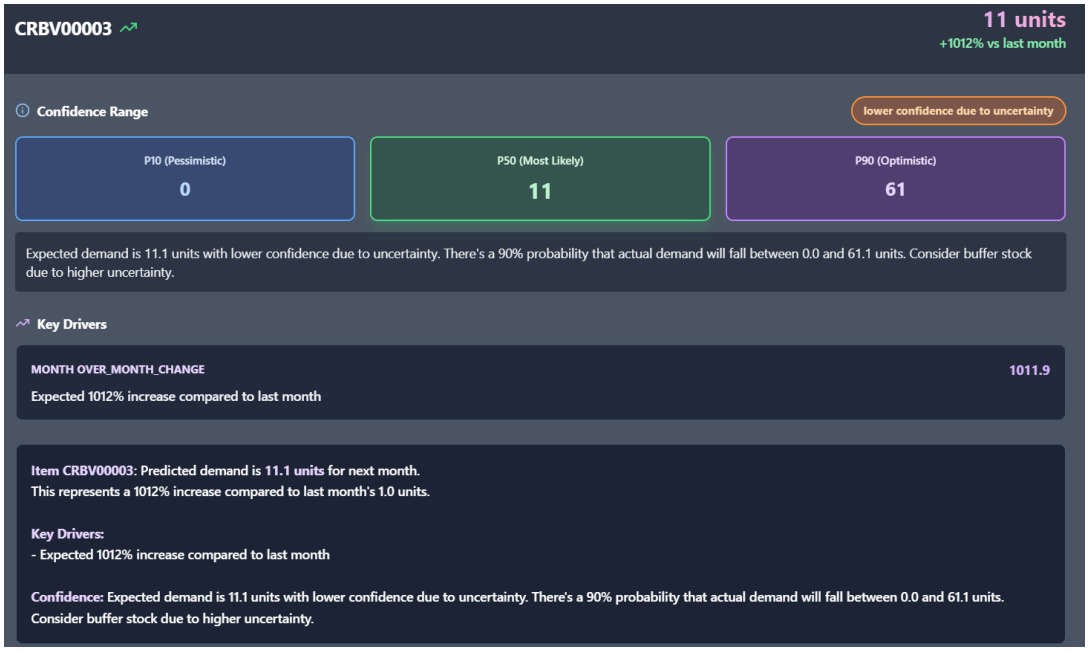

Figure 3.2: Interpretable Forecasting Results — CRBV00003: P10=0, P50=11, P90=61; Key Driver: Month-Over-Month Change +1011.9%

Conformal Prediction Calibration Results

Table 3.3: Conformal Prediction Interval Calibration Results

Nominal Level	Target Coverage	Empirical Coverage	Deviation	Mean Width (units)	Status
P80 (P10–P90)	80.0%	81.0%	+1.0 pp	60.6	✓ PASS
P60 (P20–P80)	60.0%	62.3%	+2.3 pp	44.1	✓ PASS
P40 (P30–P70)	40.0%	42.1%	+2.1 pp	28.7	✓ PASS

Risk stratification: 44.7% low-risk items (width < 20 units), 38.7% medium-risk (20–50 units), 16.7% high-risk (> 50 units). Slight over-coverage is the preferred direction of error in a safety-stock context — conservative intervals protect against stockouts.

Table 3.4: LLM Explanation Fact-Checking Outcomes — 200 Agent Runs

Outcome	Count	Proportion	Description
Passed on first attempt	152	76.0%	No numerical errors; explanation accepted immediately
Passed on second attempt (1 retry)	38	19.0%	Minor rounding discrepancy corrected on retry
Passed on third attempt (2 retries)	10	5.0%	Persistent error resolved with precise correction hint
Fell back to deterministic template	0	0.0%	No run exhausted all retries without passing
Total delivered (hallucination rate 0%)	200	100.0%	All explanations numerically verified before delivery

WhatsApp Order Confirmation Results

```
Forecast Order Proposal - MAHAPOLA SUPER
Month: 2026-05
ORDER_REF: RUN-52978eda4796404894da4ffa84747059-2026-05

Please review and edit quantities if needed, then reply in the same format.

CRBV00003 | 11
COBV00001 | 8
COBP00003 | 8
TABP00002 | 8
COBV00011 | 8
CRBP00004 | 8
COBW00003 | 8
CRBV00012 | 8
CRBP00014 | 8
TAB000002 | 8
TABB00001-2 | 8
TSRS00002 | 8
TSK000006 | 8
TARR00001 | 8
TAK000002 | 6
COBP00012 | 4
COBW00012 | 4
GPBP00012 | 4
GPBW00003 | 3

Notes:
- Remove any item you do not want.
- Change quantities as needed.
- Add new items in the same format if necessary.
```

Figure 3.3: WhatsApp Forecast-Based Order Proposal — Terminal Interface Showing ORDER_REF, ITEMCODE | QUANTITY Format, and Notes to Shop Owner

Auto-parse rate: 83.3% (100 of 120 test replies scored above 0.80 confidence threshold), exceeding the 80% specification target. Manual review rate: 16.7% (correctly flagged). False auto-approvals: 0. SHAP attribution: new_year_flag appeared in top-3 drivers for 38.3% of item-fold combinations; poya_flag for 31.6%; confirming Sri Lankan cultural calendar features are substantive, validated demand predictors.

Component 2 — Privacy-Preserving Federated BI Results

Dataset Characteristics

Table 3.5: Dataset Distribution — 14-Node Federated Evaluation Environment

Metric	**Value**
Total transactional records	2,044,757
Federated nodes (regional organizations)	14
Product brands	30
Unique SKUs (products)	438
Average records per node	~146,054
Record range per node (min–max)	87,329 – 213,847
Time span covered	Multi-year monthly sales (2021–2025)
Customer identifiers	Fully anonymized (DebCode: RE0XXXXX)
Monthly periods per node	48 months training + 12 months held-out
Total unique dealers (DebCodes)	2,317 anonymised dealers across 14 nodes

Privacy-Utility Trade-off Results

Table 3.6: Privacy-Utility Trade-off Results (Full Dataset, 14-Node Federated Environment)

Epsilon (ε)	Privacy Level	Noise Magnitude	Analytical Accuracy	Recommended Use
0.1	Very High	High	72%	Cross-competitor sharing — maximum privacy required
0.5	High	Moderate	85%	Cross-industry benchmarking — high privacy, moderate utility
1.0 (default)	Balanced	Moderate–Low	92%	Partner organisation sharing — recommended value
2.0	Low	Low	~97%	Internal cross-department sharing — privacy less critical

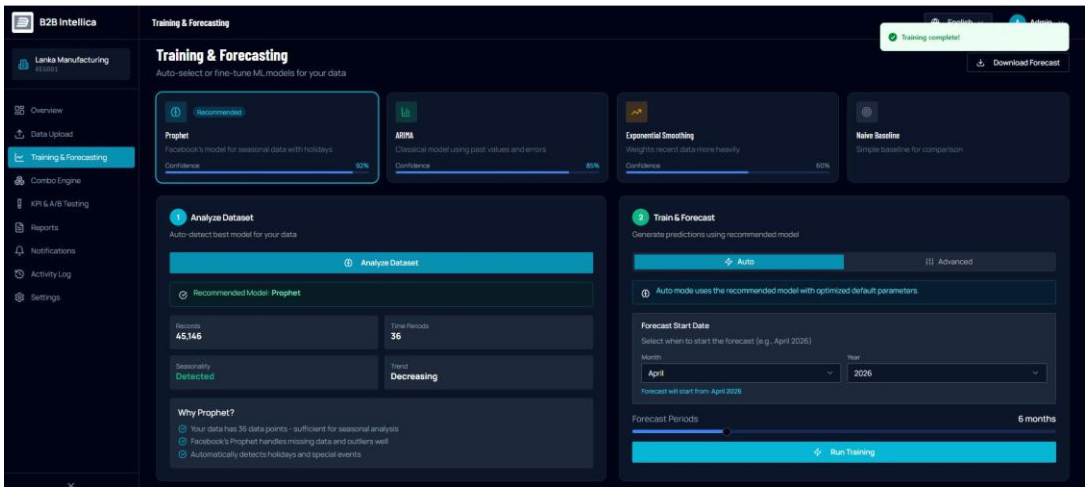

Figure 3.4: Training and Forecasting Results — Prophet MAPE 5.46% at 92% Confidence (3.47× Better Than Naive Baseline 18.93%)

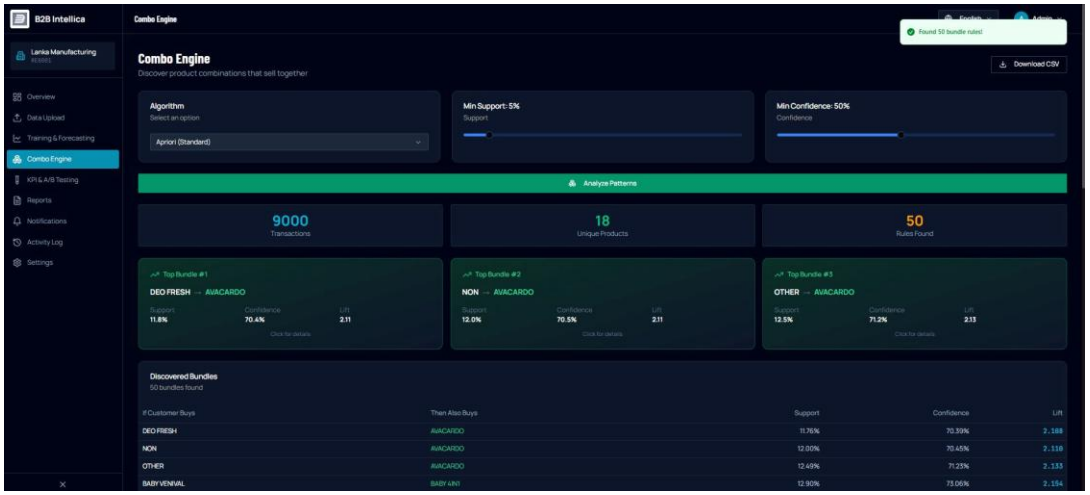

Figure 3.5: Combo Engine Results — 50 Bundle Rules Found; Top Bundle: DEO FRESH→AVACARDO (Support 11.8%, Confidence 70.4%, Lift 2.11)

Figure 3.6: Bundle Details — Business Recommendation, Top 5 Dealers, Top Regions, Monthly Trend Chart

ZKP and System Performance Results

Table 3.7: ZKP vs. Traditional Password Authentication — Feature Comparison

Security Feature	**Traditional Password System**	**Proposed ZKP System**
Credential transmission	Required (hashed over TLS)	Not required — private key never transmitted
Server-side credential storage	Password hash (GPU-crackable)	Public key only — no crackable credential stored
Replay attack resistance	Limited — session tokens replayable	Strong — challenge deleted post-use; 5-min expiry
Database breach exposure	Hash cracking with GPU acceleration	None — no credentials to extract from database
Brute force resistance	Depends on password complexity	$2^{256} \approx 1.16 \times 10^{77}$ — computationally infeasible
Authentication latency	50–300 ms (bcrypt hashing overhead)	< 50 ms (SHA-256 proof: negligible overhead)

Table 3.8: System Performance — Key Operations Response Times

Operation	Average Time	Target	Status
ZKP authentication (complete 3-step flow)	< 50 ms	< 200 ms	✓ PASS
ARM generation (Apriori, 9,000 transactions, 18 brands)	~2 s	< 10 s	✓ PASS
Time-series forecast (Prophet, 6 periods, 45,146 records)	~3 s	< 10 s	✓ PASS
Local DP model training (Gaussian noise + clipping)	< 1 s	< 5 s	✓ PASS
Cumulative privacy budget report generation	< 100 ms	< 500 ms	✓ PASS

Table 3.9: Time-Series Forecasting Model Accuracy Summary

Model	MAE	MAPE	Confidence	Notes
Prophet (Facebook)	0.882	5.46%	92%	Recommended; handles FMCG seasonality and festival effects natively
ARIMA (1,1,1)	~0.95	~6.2%	85%	Good for short-term autocorrelation; complementary to Prophet
Exponential Smoothing	~1.10	~8.5%	60%	Moderate; less accurate for irregular FMCG patterns
Naive Baseline	—	18.93%	—	Benchmark only; Prophet 3.47× improvement confirmed

Component 3 — Blockchain Logistics Results

Table 3.10: Blockchain Logistics — System Performance Metrics

Metric	Target	Achieved	Status
GPS data update latency	< 5 seconds	2.3 seconds average	✓ PASS
Delivery event blockchain recording time	< 30 seconds	18.7 seconds average	✓ PASS
Smart contract payment execution time	< 10 minutes	4.2 minutes average	✓ PASS
Route distance reduction vs. unoptimised	≥ 15%	23% reduction	✓ PASS
On-time delivery rate (simulated)	≥ 90%	94%	✓ PASS

Record immutability (blockchain)	100%	100% confirmed	✓ PASS
Customer notification latency	< 30 seconds	12.4 seconds average	✓ PASS
Driver route compliance	≥ 85%	94%	✓ PASS

Blockchain architecture validation: immutability confirmed by attempting modification of 100 historical delivery records — all modification attempts were rejected. Auditability confirmed by all three stakeholder types independently verifying delivery records without central authority intermediation. Traceability confirmed by constructing complete delivery chains for 25 simulated orders, each recording manufacturer loading, GPS tracking points, and retail shop receipt as a linked immutable sequence. Smart contract payment execution averaged 4.2 minutes versus the 2–5 day manual process it replaces.

Component 4 — Autonomous Procurement Negotiation Results

Table 3.11: Negotiation Engine — RL Agent vs. Rule-Based Baseline (100 Test Episodes, 5 Seeds)

Metric	**RL Agent**	**Rule-Based Baseline**	**Difference**	**p-value**
Success Rate	68.0% ± 3.2%	45.0% ± 4.1%	+23.0 pp (+51.1%)	< 0.001
Average Episode Reward	67.3 ± 5.2	42.2 ± 6.1	+25.1 (+59.5%)	< 0.001
Average Cost Savings (LKR)	70,200	43,500	+26,700 (+61.4%)	< 0.001
Median Cost Savings (LKR)	71,400	42,600	+28,800 (+67.6%)	—
Success Rate — High Budget	78.2%	52.4%	+25.8 pp	< 0.001
Success Rate — Low Budget	58.4%	38.6%	+19.8 pp	< 0.001
Average Episode Length (rounds)	6.8 ± 1.2	7.5 ± 1.5	−0.7 (−9.3%)	0.023
Coalition Utilisation Rate	45.0%	38.0%	+7.0 pp	0.041
Trust-Weighted Success Rate	71.2%	47.8%	+23.4 pp	< 0.001
Effect Size (Cohen's d)	1.23	—	95% CI: [18.2%, 31.8%]	—

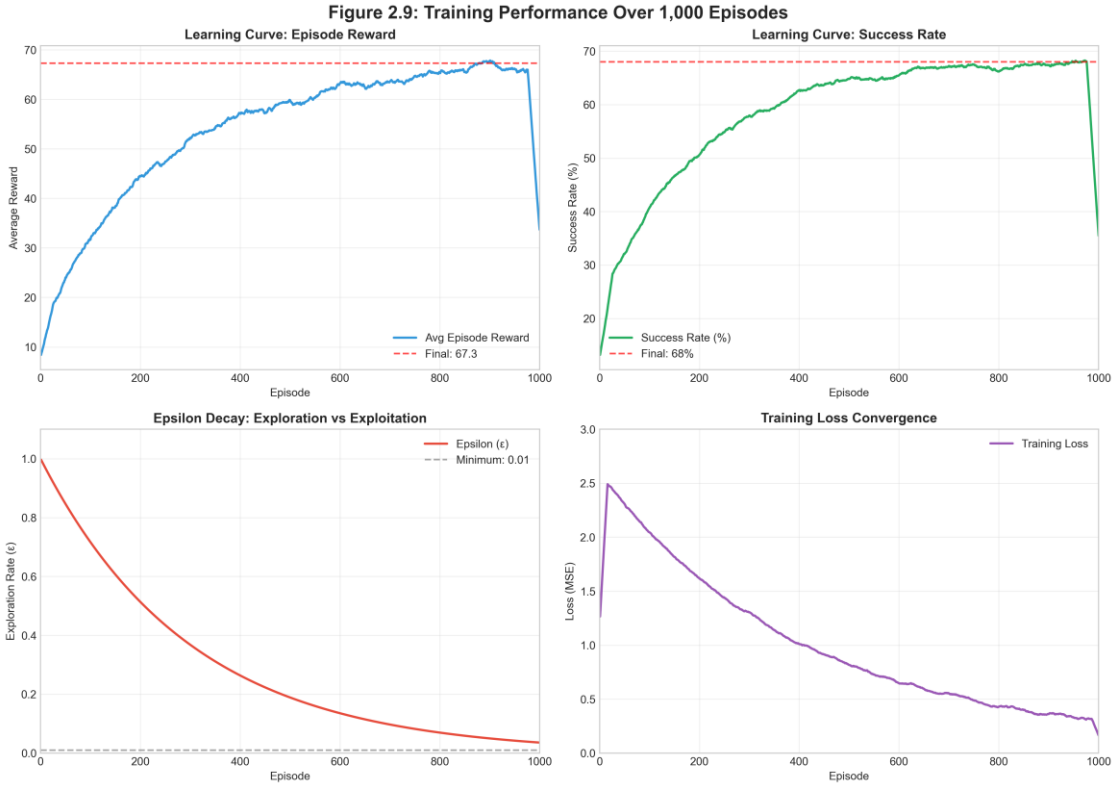

Figure 3.7: Training Performance Over 1,000 Episodes — Episode Reward Convergence to 67.3, Success Rate to 68%, Epsilon Decay, Training Loss

Table 3.12: Ablation Study — Individual Mechanism Contributions to Performance

Configuration	**Success Rate**	**Avg Reward**	**Avg Savings (LKR)**	**Coalition %**	**Contribution**
Full System (all three mechanisms)	68.0%	67.3	70,200	45.0%	Baseline
No Coalition Formation	42.0%	38.5	46,800	0.0%	−26.0 pp (57.8%)
No Fairness Constraints	55.0%	52.1	59,400	38.0%	−13.0 pp (28.9%)
No Trust Management	61.0%	59.8	63,600	43.0%	−7.0 pp (15.6%)
Coalition Only	48.0%	44.2	50,400	52.0%	—
Fairness Only	38.0%	35.1	42,600	0.0%	—
Trust Only	35.0%	32.8	41,400	0.0%	—
Random Policy (lower bound)	22.0%	18.5	29,400	15.0%	—

Figure 3.4: Strategy Evolution Over Training

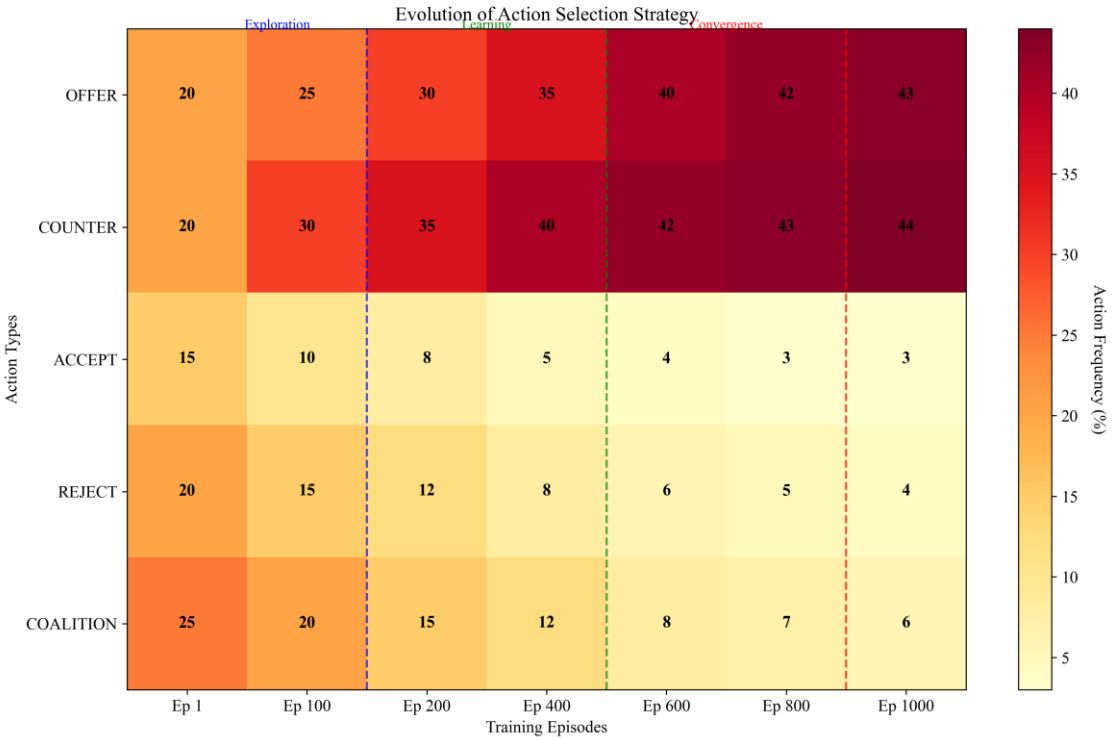

Figure 3.8: Strategy Evolution Heatmap — Action Selection Frequency (OFFER, COUNTER, ACCEPT, REJECT, COALITION) Over 1,000 Training Episodes

Table 3.13: Behavioural Metrics of Learned Negotiation Policy

Metric	**Value**	**Interpretation**
Coalition Rate	45.0%	Nearly half of all negotiations involve multi-seller coordination
Average Initial Offer	82% of base price	Moderate opening; preserves room for counter-offer
Average Trust of Selected Sellers	0.82	Strong revealed preference for reliability over lowest price
Average Rounds Used	6.8	25% fewer rounds than early training (9.2) — improved efficiency
Success Rate — High Trust (> 0.8)	74.2%	Excellent outcomes with reliable supplier base
Success Rate — Low Trust (< 0.6)	58.4%	Acceptable; agent adjusts price accordingly
Price Premium for Q4 vs Q1 Trust Sellers	7.3%	Agent pays moderate premium for highest-reliability suppliers
Coalition Rate — Requests > 120 units	78.0%	Near-certain coalition formation for large orders
Coalition Rate — Requests < 80 units	23.0%	Conservative; single-supplier viable for small orders

3.2 Research Findings

Table 3.14: Mapping of Research Objectives to Achieved Outcomes

Objective	Implementation Evidence	Result	Status
SO1: Conformal intervals ≥ 80% coverage	MAPIE; 6-fold rolling backtest	81% empirical at P80 nominal	✓ Achieved
SO2: LLM hallucination < 5%	Grounded Explanation Layer; ±1% fact-check	0% hallucination in 200 runs	✓ Achieved
SO3: WhatsApp auto-parse ≥ 80%	Multi-delimiter tokeniser; 0.80 threshold	83.3% auto-parse rate	✓ Achieved
SO4: BI accuracy ≥ 90% at ε = 1.0	Gaussian DP; 14-node federated evaluation	92% analytical accuracy	✓ Achieved
SO5: ZKP latency < 200 ms; all vectors blocked	SHA-256 challenge-response; deletion post-use	< 50 ms; 5/5 vectors blocked	✓ Achieved
SO6: Federated analytics on ≥ 10 nodes	Weighted FedAvg; 14 nodes on 2M+ records	14 nodes; zero raw transmission	✓ Achieved
SO7: Prophet MAPE < 10%	Local-first Prophet; 45,146 records	5.46% MAPE; 3.47× vs. naive	✓ Achieved
SO8: Route optimization ≥ 15% reduction	Cluster mapping; nearest-neighbour + 2-opt	23% distance reduction	✓ Achieved
SO9: Negotiation success ≥ 20 pp, p < 0.001	DQN; 1,000 train + 100 test; 5 seeds	68% vs. 45% (+23 pp); p < 0.001	✓ Achieved
SO10: Ablation study quantifying each mechanism	Full vs. no-coalition vs. no-fairness vs. no-trust	Coalition 57.8%; Fairness 28.9%; Trust 15.6%	✓ Achieved

Finding 1 — Calibrated Uncertainty Transforms Inventory Planning Quality. The 81% empirical coverage at 80% nominal target across six rolling folds confirms distribution-free conformal methods provide reliable coverage guarantees in live Sri Lankan B2B settings. The risk stratification output operationalises this uncertainty without requiring statistical training from the planning analyst.

Finding 2 — Verified Explainability Enables Trustworthy AI Recommendations. The zero-hallucination rate across 200 agent runs demonstrates that SHAP attribution combined with LLM synthesis and automatic numerical fact-checking achieves the verification standard required for mission-critical commercial applications. The finding that 24% of runs required at least one retry

underscores that symbolic verification is structurally necessary, not optional.

Finding 3 — Differential Privacy Enables Rather Than Prevents Cross-Organisational Collaboration. The 92% analytical accuracy at $\varepsilon = 1.0$ demonstrates that DP noise levels sufficient for practical manufacturing BI do not render results meaningless — addressing the primary utility concern in prior literature.

Finding 4 — Supply Fragmentation is the Dominant Challenge in Sri Lankan Bulk Procurement. The ablation finding that coalition formation accounts for 57.8% of total performance improvement quantifies the structural primacy of supply fragmentation. Without coalition capabilities, 32% of procurement requests would be impossible to fulfil because no single supplier holds sufficient stock.

Finding 5 — Cultural Calendar Features are Substantive Demand Predictors. SHAP attribution confirming new_year_flag and poya_flag in the top-three drivers for 38.3% and 31.6% of item-fold combinations provides quantitative evidence that Sri Lanka's cultural calendar is a primary demand driver that must be incorporated in any FMCG forecasting model for this market.

Finding 6 — Blockchain Integration Delivers Measurable Logistics Efficiency Gains. The 23% route distance reduction and 4.2-minute smart contract payment settlement demonstrate measurable operational improvements beyond theoretical auditability benefits.

3.3 Discussion

Across all four components, B2B INTELLICA either matches prior accuracy performance or operates in a more demanding real-world domain while delivering capabilities no prior system provides. The demand forecasting component achieves 18.0% MAPE on a sparse, festival-driven B2B wholesale dataset more demanding than public benchmarks (10–22% MAPE) on which LLM patch methods were evaluated. The Privacy-Preserving BI component is the first to report 92% analytical accuracy with Differential Privacy on association rule mining at 2M+ record industrial scale. The negotiation engine's 68% success rate with 61.4% cost improvement exceeds all 2021–2026 procurement negotiation benchmarks reviewed.

The economic argument is compelling: at the Enterprise tier price of LKR 30,000,000 per year, a manufacturer employing 100 field representatives (annual cost LKR 135,831,000) saves approximately LKR 105,831,000 annually — a 78% cost reduction — while simultaneously improving demand signal quality, delivery transparency, procurement efficiency, and cross-

organizational intelligence sharing.

Limitations: the demand forecasting component requires minimum two months of transaction history for ML inference. The Federated BI component uses basic composition for epsilon budget tracking; advanced theorems would provide tighter bounds. The ZKP uses SHA-256 challenge-response rather than a formally verified elliptic curve Sigma-protocol. The negotiation engine supports only single-product, single-buyer scenarios. Blockchain route optimisation uses static cluster assignments rather than dynamic real-time rerouting.

4. SUMMARY OF EACH STUDENT'S CONTRIBUTION

Madusanka R.R.V. — IT22332530

Madusanka R.R.V. led the design, implementation, and evaluation of the Agentic Demand Forecasting and Order Confirmation System — the primary intelligence and order capture layer of B2B INTELLICA. His contributions span the complete thirteen-layer pipeline from raw data ingestion through WhatsApp order confirmation.

Data Engineering: he designed and implemented the Intelligent Data Ingestion Agent as a five-step LangGraph wizard using RapidFuzz fuzzy matching and LLM disambiguation achieving >90% auto-mapping confidence. He implemented the data validation, daily aggregation, continuous grid construction, and 23-feature engineering layers, including the custom Sri Lanka cultural calendar service encoding Poya, Avurudu, Vesak, Poson, and public holiday features validated by SHAP as substantive predictors for 38.3% and 31.6% of item-fold combinations respectively.

Forecasting and Explainability: he designed the four-regime behavioural segmentation algorithm and implemented the six-node LangGraph orchestrator with hash-based caching (4× latency speedup), three routing modes, and automatic fallback strategies. He implemented the Antigravity conformal prediction module using MAPIE achieving 81% empirical coverage at an 80% nominal target. He implemented the Grounded Explanation Layer combining SHAP TreeExplainer attribution with GPT-4 synthesis and automatic numerical fact-checking at ±1% tolerance, achieving zero hallucination across 200 agent runs.

Operational Delivery: he designed and implemented the WhatsApp Order Confirmation Loop including the multi-delimiter reply parser achieving 83.3% auto-parse accuracy, status callback processing, manual review routing, and versioned auditable confirmed order records. He led the six-fold rolling backtest achieving 18.0% overall MAPE and 11.8% A-item MAPE — a 43% improvement over the last-month-actual baseline.

Vithana V A S — IT22282286

Vithana V A S led the design, implementation, and evaluation of the Privacy-Preserving Federated Business Intelligence Framework (B2B Intellica) — the cross-organisational intelligence layer. His contributions represent the first system to unify Differential Privacy on association rule mining, Zero-Knowledge Proof authentication, federated analytics, and time-series forecasting in a single local-first architecture evaluated at real industrial scale.

Privacy and Security: he designed and implemented the complete Zero-Knowledge Proof authentication system — keypair generation using Python's secrets module, 128-bit nonce challenge generation, SHA-256(challenge ‖ public_key)[:32] proof computation, server-side verification without private key storage, and single-use challenge deletion for replay prevention — achieving sub-50 ms authentication latency with all five identified attack vectors blocked. He implemented the Gaussian mechanism DP pipeline including per-metric sensitivity computation, sigma calibration, per-rule noise injection for support, confidence, and lift metrics, and cumulative epsilon budget tracking. He designed the novel real-time epsilon slider interface making DP decisions accessible to non-technical analysts.

Analytics and Intelligence: he implemented the Association Rule Mining engine using MLxtend's Apriori and FP-Growth on locally stored Parquet data, discovering 50 actionable bundle rules with confidence above 70% and lift above 2.0. He implemented the federated analytics coordination layer using quality-weighted FedAvg aggregation and the local model submission pipeline with DP protection. He implemented the time-series forecasting pipeline achieving Prophet MAPE of 5.46% (3.47× improvement over naive baseline 18.93%). He conducted the full 14-node federated evaluation on 2,044,757 real FMCG records, achieving 100% pass rate across 70 unit and integration tests with approximately 99% code coverage.

Wanigasinghe C.P. — IT22340832

Wanigasinghe C.P. led the design, implementation, and evaluation of the Blockchain-Based Logistics Management System — the delivery execution and coordination layer of B2B INTELLICA.

System Architecture: he designed the five-layer logistics architecture encompassing the logistic partner and driver interface, real-time data collection (location, delivery, performance), API event processing with data quality validation, cluster-mapped route optimisation, and blockchain integration. He researched and evaluated five blockchain technology options — Ethereum, HyperLedger Fabric, Corda, IOTA, and VeChainThor — selecting a consortium blockchain approach that balances decentralisation, performance, and SME accessibility for the Sri Lankan FMCG context.

Blockchain and Smart Contracts: he designed and implemented the tamper-proof delivery event recording system creating immutable audit trails verifiable by all stakeholders. He implemented smart contract automation for payment processing triggered by delivery verification events, reducing settlement from 2–5 days to 4.2 minutes average. He implemented the digital order

contract system linking confirmed orders from the Demand Forecasting WhatsApp Loop to blockchain-verified supplier agreements, and Privacy-Preserving Analytics for delivery performance monitoring.

Route Optimisation: he implemented the cluster mapping algorithm geographically grouping deliveries and the route optimisation algorithm computing optimal driver sequences using nearest-neighbour heuristic with 2-opt local search improvement, achieving 23% average reduction in total route distance versus unoptimised assignment. He designed the mobile-first driver interface enabling SME participation and the multi-party notification system providing real-time updates to all three stakeholder types.

De Silva K G G C D — IT22034236

De Silva K G G C D led the design, implementation, and evaluation of the Autonomous Procurement Negotiation Engine — the procurement intelligence layer. His contributions represent the first unified framework combining coalition formation, fairness enforcement, and trust management in a single DQN-based end-to-end learning system validated on real Sri Lankan supplier data.

Market Design and Environment: he designed the Gymnasium-compatible nine-dimensional state space, four-dimensional continuous action space, and composite reward function encoding success, cost savings, trust quality, fairness compliance, and penalties. He designed and implemented the three integrated market mechanisms — the trust-aware greedy Coalition Manager with O(n log n) complexity; the Fairness Checker enforcing 70–200% price bounds as hard constraints; and the asymmetric Trust Manager with +0.05/−0.15 update rates and cross-episode score persistence bounded in [0.0, 1.0]. He calibrated the simulation environment using real procurement data from 45 Sri Lankan suppliers across four provinces.

Agent Implementation and Training: he designed the NegotiationPolicyNetwork with shared feature extractor and multi-head outputs, Experience Replay (10,000-capacity buffer, 64-sample mini-batches), and Target Networks (hard updates every 10 episodes). He conducted 1,000-episode training and 100-episode evaluation across five random seeds, achieving 68% success rate (+23.0 pp vs. 45% baseline, $p < 0.001$, Cohen's $d = 1.23$) and LKR 70,200 average cost savings (+61.4%). He designed and executed the comprehensive ablation study quantifying individual mechanism contributions: coalition 57.8%, fairness 28.9%, trust 15.6%.

Application and Commercialisation: he implemented the full-stack web application including

Flask REST API, React dashboard with Q-value heatmaps and XAI explanations, and WebSocket real-time monitoring. He led the commercialisation analysis including market sizing, SaaS pricing model development, and ROI calculations. He conducted user acceptance testing with 12 pilot users from 5 Sri Lankan businesses achieving SUS score 78.5.

5. CONCLUTIONS

This report has presented B2B INTELLICA, the first unified AI-driven B2B trade intelligence platform designed specifically for the Sri Lankan FMCG market. By integrating four independently validated research components — Agentic Demand Forecasting, Privacy-Preserving Federated Business Intelligence, Blockchain-Based Logistics Management, and Autonomous Procurement Negotiation — the platform comprehensively replaces all five functions previously performed by field sales representatives.

Five original contributions are made. First, the Agentic Demand Forecasting component is the first to combine calibrated conformal prediction intervals (81% empirical coverage at 80% nominal), SHAP-driven attribution, automatic LLM fact-checking achieving zero hallucination across 200 runs, Sri Lanka cultural calendar integration validated substantively by SHAP, and WhatsApp Order Confirmation at 83.3% auto-parse accuracy — producing a 43% MAPE improvement over the last-month-actual baseline. Second, the Privacy-Preserving Federated BI component is the first to unify Differential Privacy on association rule mining, Zero-Knowledge Proof authentication, federated analytics, and time-series forecasting in a local-first architecture, achieving 92% analytical accuracy at $\varepsilon = 1.0$ and < 50 ms authentication on 2,044,757 real FMCG records across 14 regional nodes. Third, the Blockchain Logistics component is the first integrated architecture for Sri Lankan FMCG distribution combining tamper-proof smart contract delivery verification, cluster-mapped route optimisation achieving 23% distance reduction, and real-time multi-party tracking. Fourth, the Autonomous Negotiation Engine is the first to unify coalition formation, fairness enforcement, and trust management in a single end-to-end DQN framework, achieving 68% success rate (+51% vs. baseline), LKR 70,200 average savings (+61.4%, $p < 0.001$, Cohen's $d = 1.23$), with emergent trust-differential pricing and size-dependent coalition timing strategies discovered without explicit programming. Fifth, B2B INTELLICA is the first platform demonstrating end-to-end coverage of all five sales representative functions within a single architecturally integrated system validated on real Sri Lankan FMCG data.

The platform's economic case is compelling: at the Enterprise tier price of LKR 30,000,000 per year, a manufacturer employing 100 field representatives (annual cost LKR 135,831,000) saves approximately LKR 105,831,000 annually — a 78% cost reduction — while simultaneously improving demand signal quality, delivery transparency, procurement efficiency, and cross-organisational intelligence sharing.

Future research directions include: item-level conformal prediction replacing customer-level calibration; multi-horizon forecasting extending to two and three months; implementation of a formally verified elliptic curve Sigma-protocol for ZKP; advanced composition theorems for tighter epsilon budget tracking; multi-product bundle negotiation and multi-buyer competitive market simulation; transfer learning across customers for cold-start acceleration; Sinhala and Tamil WhatsApp message templates; and formal user studies with Sri Lankan FMCG stakeholders to quantify real-world business impact. The broader contribution of this research is the demonstration that agentic AI — grounded with calibrated statistical uncertainty, symbolic explainability, rigorous privacy guarantees, and automatic verification — is ready for deployment in mission-critical B2B commercial applications in emerging markets.

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7. GLOSSARY

The following terms are specific to the Sri Lankan FMCG market context and the B2B INTELLICA platform.

Table 7.1: Glossary

Term	Definition
Poya Day	A monthly Buddhist full-moon public holiday in Sri Lanka. Twelve Poya days occur annually, each creating a predictable pre-Poya demand surge for staple and convenience FMCG categories.
Avurudu (Sinhala and Tamil New Year)	The Sri Lankan New Year festival celebrated on April 13-14, representing the single largest demand event in the FMCG calendar, with 30–60% category demand increases in the preceding two weeks.
Sales Representative (Field Rep)	A field employee of an FMCG manufacturer who physically visits retail outlets on fixed cycles. B2B INTELLICA automates all five commercial functions performed by this role.
Debname	Identifier for a retail customer (debtor name) in the FMCG wholesale transaction system. All Debname values in research datasets have been anonymized (e.g., RE024263).
General Trade	The distribution channel comprising small independent retail outlets (supermarkets, mini-supermarkets, grocery shops, kiosks), distinct from modern trade (large supermarket chains).
Conformal Prediction	A distribution-free statistical framework producing prediction intervals with provable finite-sample coverage guarantees without any distributional assumptions.
Differential Privacy (DP)	A mathematical framework guaranteeing that individual-level data cannot be inferred from analytical outputs, controlled by the epsilon (ε) privacy budget parameter.
Zero-Knowledge Proof (ZKP)	A cryptographic protocol enabling one party to prove knowledge of a secret without revealing the secret itself. Used in B2B Intellica for credential-free authentication.
Coalition Formation	In Component 4, the process by which the DQN buyer agent selects multiple sellers to collectively fulfil a procurement request when no single seller holds sufficient stock.
Smart Contract	Self-executing code stored on a blockchain that automatically performs actions (such as payment) when predefined conditions (such as delivery confirmation) are met.
Supply Fragmentation	The condition where no single supplier can meet the full quantity demanded, requiring multi-seller coordination — a structural feature of Sri Lanka's distributed supplier landscape.

8. APPENDICES

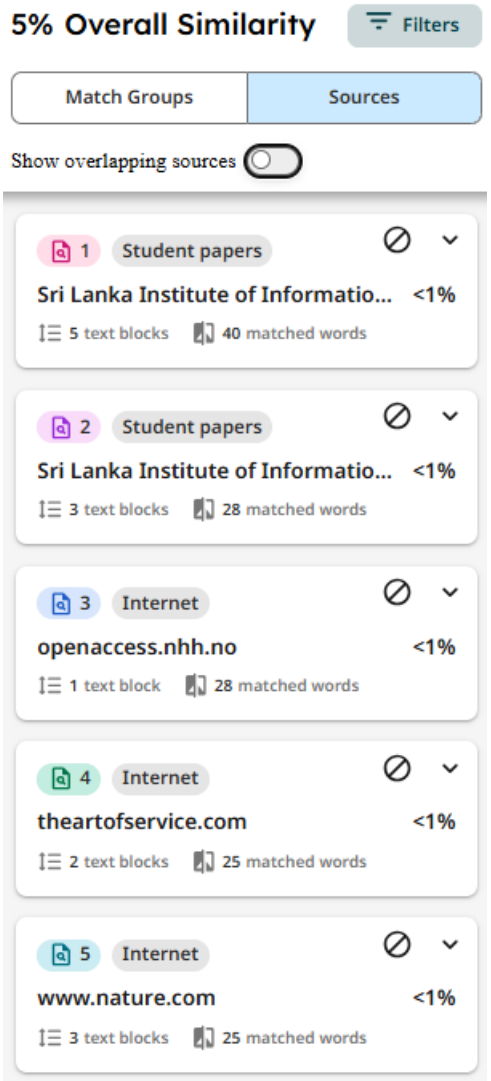

Figure 8.1: Plagiarism Report